

Gisma University of Applied Sciences

HUMAN FACTORS IN CYBERSECURITY: THE ROLE OF EMPLOYEE AWARENESS IN REDUCING CYBER THREATS

Name : Akash Upadhyay

ID : GH1048531

PPT : GitHub

ABSTRACT

Cybersecurity vulnerability is still a major concern to organisations and employee behaviour is a major cause of cybersecurity vulnerability. In addressing the issue of how employee awareness can be used to curb cyber threats, this dissertation summarizes the academic literature available on human factors in cybersecurity. The research article has adopted an interpretivist philosophy, inductive research method and a qualitative secondary research design based on a systematic literature review. The correlation between employee awareness, cybersecurity behaviour and organisational security was investigated by identifying and thematically analysing the appropriate literature.

The analysis has highlighted three key themes, i.e. human error as the most common cybersecurity risk, programmemes of employee awareness, and culture of security in the organisation. The results suggest that phishing, social engineering, poor use of passwords, negligence, insider threat and security fatigue have the potential to add organisation vulnerability. Awareness programmes have the potential to enhance employee knowledge and security practices, however, no-one can be sure that this will be a permanent behaviour. Ongoing, hands-on and quantifiable training would thus be more suitable as compared to the annual awareness programs. The results also indicate that leadership, communication, adherence to policies and organisational accountability are some of the contributors of transformation of cybersecurity awareness to secure employee behaviour.

The research concludes that employee awareness is an effective basis of cybersecurity and secure behaviour is the process based on which awareness can translate into cyber threat mitigation. This research recommends constant awareness training, realistic simulation, measurement of behaviour, efficient reporting system and participation of leaders and positive organisational culture of security. Throughout the dissertation, it is typically shown that a humanistic and organisational methodology involving the incorporation of employee awareness and behavioural practices, organisational culture and proper technological controls are required to neutralize cyber threats.

Table of Contents

Chapter 1: Introduction 7

1.1 Background of the Study 7

1.2 Research Context 7

1.3 Research Problem 8

1.4 Research Aim 8

1.5 Research Objectives 8

1.6 Research Questions 9

1.7 Significance of the Study 9

1.8 Scope and Delimitations 10

1.9 Dissertation Structure 10

Chapter 2: Literature Review 11

2.1 Introduction 11

2.2 Understanding Cybersecurity 11

2.3 Human Factors in Cybersecurity 13

2.4 Employee Awareness and Security Culture 16

2.5 Cybersecurity Awareness Programmes 18

2.6 Theoretical Framework 19

2.6.1 Protection Motivation Theory 19

2.6.2 Theory of Planned Behaviour 21

2.7 Literature Gap 22

2.8 Conceptual Framework 23

Chapter 3: Methodology 24

3.1 Introduction 24

3.2 Research Philosophy 24

3.3 Research Approach 24

3.4 Research Design 25
3.5 Research Strategy: Systematic Literature Review 26
3.6 Inclusion and Exclusion Criteria 26
3.7 Literature Search and Selection Process 26
3.8 Data Processing and Keyword-Based Classification 27
3.9 Software Design and Implementation 27
3.9.1 Software Architecture and Project Structure 27
3.9.2 Data Cleaning and Literature Validation 29
3.9.3 Keyword-Based Coding and Manual Verification 29
3.9.4 Verified Analysis, Visualisation and Output Generation 29
3.10 Data Extraction Framework 29
3.11 Trustworthiness and Quality Assurance 30
3.12 Ethical Considerations 30
3.13 Methodological and Software Limitations 30
3.14 Chapter Summary 31
Chapter 4: Analysis and Findings 32
4.1 Introduction 32
4.2 Research Dataset and Output Overview 32
4.3 Literature Validation and Data Cleaning Results 33
4.4 Candidate Classification Results 34
4.5 Manual Review and Verification Results 35
4.6 Verified Classification and Summary Results 35
4.7 Visualisation and Dashboard Output Analysis 37
4.8 Analysis of the Three Predefined Research Categories 38
4.8.1 Category 1: Human Error and Cybersecurity Risk 38
4.8.2 Category 2: Employee Awareness Programmes 39

4.8.3 Category 3: Organisational Security Culture 40
4.9 Evaluation of the Software Workflow 41
4.10 Findings in Relation to Research Objectives 41
4.11 Chapter Summary 42
Chapter 5: Discussion 43
5.1 Introduction 43
5.2 Discussion of Research Question 1: Employee Awareness and Organisational Cybersecurity 43
5.3 Discussion of Research Question 2: Human Factors Behind Cybersecurity Threats 44
5.4 Discussion of Research Question 3: Effectiveness of Awareness Programmes 45
5.5 Discussion of Research Question 4: Measures to Improve Employee Awareness 46
5.6 Theoretical Interpretation Using Protection Motivation Theory and Theory of Planned Behaviour 47
Chapter 6: Conclusion 49
6.1 Introduction 49
6.2 Achievement of the Research Aim and Objectives 49
6.3 Summary of Key Findings 50
6.4 Contributions to Knowledge and Software Development 50
6.5 Research and Software Limitations 51
6.6 Recommendations 52
6.7 Future Research 52
References 53
Appendix 60

List of Figures

Figure 2.1: Human Factors in Cybersecurity 13
Figure 2.2: Employee Awareness and Security Culture 16
Figure 2.3: Protection Motivation Theory 20
Figure 2.4: Theory of Planned Behaviour 21
Figure 2.5: Conceptual Framework 23
Figure 3.1: Research Approach 25
Figure 3.2: Python-Based Research Software Workflow 28
Figure 4.1: Candidate Classification Results 33
Figure 4.2: Verified Classification Results 36
Figure 4.3: Cybersecurity Awareness Research Dashboard 38

Chapter 1: Introduction

1.1 Background of the Study

Digital transformation has drastically changed how the organisations are going to operate, communicate and deliver products and services. Among other things, the digital infrastructure, cloud applications, connected information systems and online communication channels are becoming increasingly dependent upon by organisations to help them carry out business as usual processes. Although these developments offer a chance of efficiency, flexibility and innovation, they also put organisations at risk of cybersecurity attacks.

The cybersecurity landscape has been an increasingly complex environment due to organisations continually being threatened by phishing, ransomware, malware, social engineering and business email compromise. Human dimension has thus emerged as a significant focus of research on cybersecurity. Recent research indicates that organisations need to scrutinise the actions of their workforces, along with technological controls, as they strive to become more cybersecure. In a systematic review of the cybersecurity training methods, the results showed that most of the studies reviewed had positive outcomes of training, yet the data on behavioural change and the most efficient methods of training was inconsistent (Duzenci, Kitapci and Gok, 2023).

1.2 Research Context

The present companies are extremely reliant on information technology as a means of communication, data management, financial transactions, customer services and internal operations. This dependency means that a cybersecurity incident can have consequences beyond the loss of information that can affect business continuity and organisational performance.

Phishing is of particular interest as it tries to use human judgement and not necessarily bypass technical security measures. The employees may be subjected to fraud emails, messages or websites that appear real and ask them to give out information or indulge in unsafe activities (Edeh, 2023). The NIST recommendations are geared towards guaranteeing that the employees should be able to spot a phishing attack and report on the occasion that they suspect that they may be under attack in addition to the importance of employee training and controls that are resistant to phishing attacks, particularly multi-factor authentication (nist.gov, 2026).

1.3 Research Problem

Although organisations have extensively invested in cybersecurity technologies, they still face the risk of threats due to human behaviour. Some of these risks can be addressed under technical means such as firewalls, anti-virus programs, access controls and authentication controls, yet they will never be able to eliminate the risk of decisions being made by employees that increase the vulnerability of security (Jones, 2024). The recurrent importance of phishing, social engineering, credentials and other types of threats usage demonstrates that cybersecurity cannot be seen only through the technological prism, however, it should be viewed through the human aspect prism.

Among the most important problems, there is the fact that when an employee is aware, it does not necessarily translate to secure behaviour. Simple rules of cybersecurity may be familiar to workers, but they may not apply these rules in practice or when faced with effective work or social engineering campaigns. Research literature on cybersecurity training has determined some mostly positive effects on knowledge and attitude, with less consensus on the potential to modify behaviours in the long-term (Khadka and Ullah, 2025). An example of a 2024 meta-analysis did show a greater overall effect of training on knowledge and attitudes compared to actual behaviour, meaning that awareness alone is not always necessary to generate various lasting behavioural changes (guides.himmelfarb.gwu.edu, 2026).

As a result, the need to summarize the existing scholarly knowledge on the relationship between employee awareness and organisational cybersecurity arises. Human factors that drive cyber threats should also be identified and the use of some awareness methods should be crucial (Kioskli *et al.*, 2023). Knowledge of such concerns can assist organisations to surpass compliance-based training towards strategies that aid secure behaviour and enhance cybersecurity cultures.

1.4 Research Aim

This paper will be dedicated to the significance of employee awareness in the reduction of cyber threats by researching the existing scholarly literature on human factors in cybersecurity.

1.5 Research Objectives

The four goals of the study are as follows:

1. To investigate the connection between employee awareness and organisational cybersecurity.
2. To locate typical human factors that could lead to cybersecurity threats.
3. To identify the effectiveness of cybersecurity awareness programmes registered in previous studies.
4. To offer suggestions on the measures that organisations can adopt to raise awareness among employees and reduce cybersecurity threat.

1.6 Research Questions

Directly based on the research aim and objectives, the dissertation is going to answer the following research questions:

RQ1: How is employee awareness related to organisational cybersecurity?

RQ2: What are the usual human aspects that can lead to cybersecurity threats within organisations?

RQ3: To what extent do cybersecurity awareness programmes increase employee cybersecurity behaviour and lessen cyber threats?

RQ4: Which are the actions that organisations can take to raise the awareness of their employees and reduce cybersecurity threats?

1.7 Significance of the Study

The research is important since cybersecurity is an organisational effort that goes beyond the application of technical controls. The human aspect of cybersecurity can be applied to businesses to come up with more effective strategies for countering inevitable security threats. Rather than looking at employees as potential source of vulnerability, organisations can use awareness programs to develop employees as a useful organisational defence (Ejiofor *et al.*, 2025).

The research also applies to the employees as it highlights the significance of personal choice to ensure safety of the organisational information and systems. A higher level of awareness will help employees to be more aware of suspicious activity, expectations of organisational security, and react to potential threats (Georgiadou, Mouzakitis and Askounis, 2022).

The study provides a summary of available evidence to the cybersecurity managers in terms of human factors, awareness programmes and security culture. This can guide the decision in training design, regular occurrence of programmes, behavioural appraisal and overall organisational creation plans. The most recent NIST recommendations also suggest a lifecycle of cybersecurity and privacy education that fosters behaviour change and assists in creating a culture of security and privacy (Hakimi, Quchi and Fazil, 2024).

The study has scholarly value uniting the last-minute literature on employee awareness, human behaviour, cybersecurity training, and organisational security culture. Recent data indicates that the management needs to be involved in cybersecurity culture, employees need to understand it, policies need to be adhered to, and staff should receive security training and dedication to the practice instead of being a technical issue (Hijji and Alam, 2022).

1.8 Scope and Delimitations

The dissertation is sensitive to the contribution of human factors and awareness of employees towards cybersecurity of organisations. It follows a secondary qualitative research design and analyses academic literature instead of gathering data on employees or organisations form primary sources. The literature review will consider the articles published in the last two to six years (2022 to 2026) and the corresponding research is peer reviewed and written in English and attainable to employee awareness, human behaviour, cybersecurity training and organisational security (Jabir, Le and Nguyen, 2025). Technical cybersecurity solutions, software development or network engineering are not the focus of the research. It is quite humane and organisational-based and emphasises what can be done to ensure safer behaviour of employees via cybersecurity.

1.9 Dissertation Structure

The dissertation is written into six chapters. Chapter 1 of the research presents the background, context, problem, aim, objectives, research questions, significance and scope. Chapter 2 critically examines the literature on cybersecurity and human factors, employee awareness, security culture, awareness programmes and theoretical perspectives of relevance and then defines the literature gap and conceptual framework. Chapter 3 reports about secondary qualitative, systematic literature review, PRISMA selection process and thematic analysis. The findings are reported in chapter 4 along three important themes. Chapter 5 presents the results through the

prism of the research questions and theoretical framework and comments on the viability as a business implication. Chapter 6 summarizes the dissertation by looking at the success of the aim, highlights of the findings, contributions, limitations, recommendations and future research.

Chapter 2: Literature Review

2.1 Introduction

Conventionally, cybersecurity was associated with technical controls, e.g., network protection, access control, authentication and malware detection. However, the increasingly sophisticated nature of cyber threats has demonstrated that human behaviour is a significant factor that defines the security of organisations as well. Employees are in touch with the information systems, emails, applications and organisational data at any given time, and therefore, their awareness, decision-making and compliance with security practices can determine the exposure of the organisation to cyber risks.

The chapter consists of a literature review of the role of employee awareness in reducing cyber threats which is critical. It starts by looking at the definition and organisational significance of cybersecurity and then proceeds to look at the human aspect that leads to cyber vulnerabilities. The following chapter evaluates the awareness of employees, security culture and cybersecurity awareness programmes, and phishing simulation, in particular, password management education, gamification, training frequency and how to measure the effectiveness of the programmes. Theory of Planned Behaviour (TPB) and Protection Motivation Theory (PMT) may be considered as theoretical methods of studying employee cybersecurity behaviour. The chapter then identifies gaps in the existing body of literature and develops a conceptual model between employee awareness and secure employee behaviour and minimised cyber threats.

2.2 Understanding Cybersecurity

Cybersecurity has been a crucial organisational need due to the growing reliance on digital technologies as the variety of systems, information, and activities needing protection has grown. The way communication, storage of data, financial operations, customer management and operation processes are carried out in organisations has been interconnected information systems. This increased dependency creates opportunities to create cyber-threats to not only avert technical infrastructure but organisational activities and people as well. Aljumaiah *et al.* (2025)

describe the fact that cybersecurity risks might be observed in various elements related to IT infrastructure and the role of a logical risk identification and management plan relying on well-developed cybersecurity frameworks.

Cybersecurity, in this sense, can be considered a multidimensional organisational process, which involves the protection of information, systems, networks and digital resources against unauthorised access, interference, alteration, or abuse. Safitra, Lubis and Fakhrurroja (2023) posit that current cyber threats require a comprehensive approach whereby any organisation should always come up with new approaches to intrude into the digital environment because attackers are continuously developing their tactics. This implies that organisations cannot rely on one security measure to mitigate cybersecurity threats.

Malware, phishing, ransomware, social engineering, credential compromise, insider threats and data breaches are some of the threats that are present in the cybersecurity environment. The threats too have become more complex with the advent of digital technologies as employees contact a myriad of systems and communication channels. Qureshi and Koo (2026) illustrate the scope of modern cybersecurity issues by emphasizing the interplay between cyber threats and data privacy threats in intricate healthcare information systems.

The conventional focus on the protection of technologies has henceforth become inadequate. The way people communicate with organisational systems and security processes is also a key to cybersecurity. Edeh (2023) identifies human factors to be a notable element of cybersecurity and explains that human behaviour can influence its effectiveness as technical security control.

The importance of the employees is particularly self-evident when it is pointed out that many cyber threats are directed towards affecting users, rather than directly defeating technical protection. The complexities of phishing attacks related to human-targeting are increasing, as Jabir, Le and Nguyen (2025) demonstrate, and human factors are a vital variable to take into consideration about the vulnerability of cybersecurity.

Cybersecurity is thus a blend of technological, human and organisational factors that organisations must take into consideration. Al-Nuaimi (2024) argues that human and contextual factors are also major determinants of organisational cybersecurity, implying that cybersecurity outcomes depend on the correlation between people and their organisational context.

2.3 Human Factors in Cybersecurity

Human factors: These characteristics are the attributes, behaviours, decisions, attitudes and situations that influence the way in which people are related to the systems and practices of cybersecurity. In the organisational setting it can be possible to increase the level of cybersecurity by doing the right thing as well as creating possible holes through errors, erroneous decisions or nonadherence to security guidelines.

Alsharida *et al.* (2023) demonstrate that human cybersecurity behaviour is contingent upon the different perspectives and not on a behavioural factor. Their review of the literature highlights the importance of considering the different personal and situational factors affecting cybersecurity behaviour.

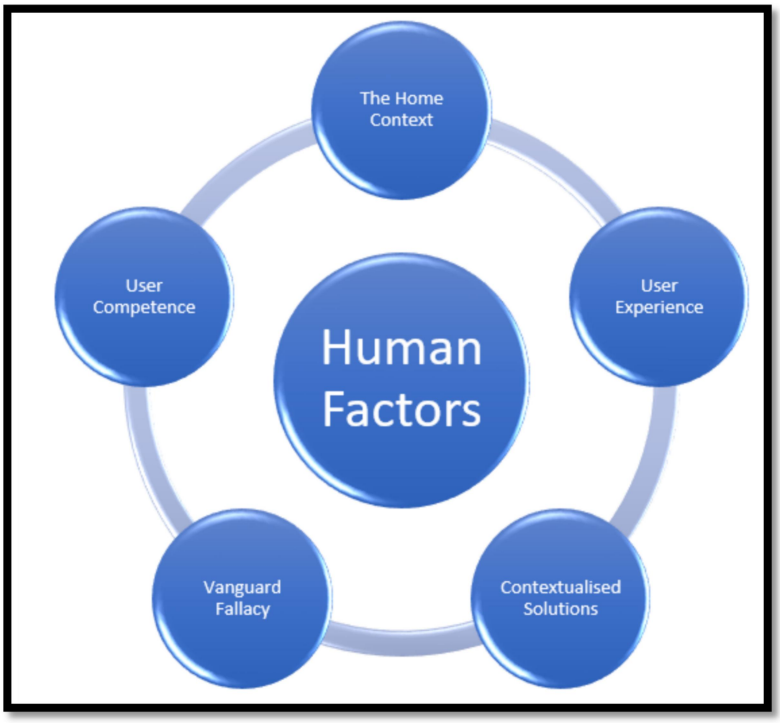


Figure 2.1: Human Factors in Cybersecurity

(Source: cyberireland.ie, 2026)

Human Error

Human error is one of the most significant human-related cybersecurity issues. By clicking on malicious links, sending sensitive information, using improper passwords, not updating their

systems or improperly dealing with organisational data, employees may unintentionally create vulnerabilities.

NcubeKuzi (2022) mentions human error as a significant cybersecurity issue with small businesses and outlines employees as a key potential vulnerability in organisational cybersecurity. This is important as smaller organisations might lack resources to employ elaborate security measures and employee awareness and behaviour become more important.

Stress and fatigue can also influence the behaviour of employees. Nobles (2022) lists stress, burnout and fatigue in the security of information as some of the human-related challenges in cybersecurity, meaning that employees have an opportunity to lose interest in security requirements under significant mental or organisational pressure.

This implies that organisations should be aware of the circumstances that cause human error instead of just apportioning blame on employees following an incident. Tikanmaki and Ruoslahti (2024) emphasize the need to comprehend human factors within the framework of the process of ensuring cybersecurity, which contributes to the importance of human-centred solutions.

Phishing and Social Engineering

Phishing and social engineering are important examples of human-centric cyber threats, because they are aimed at influencing the decision making of the employees. Some of the email messages that can be sent to employees will appear legitimate, messages or requests that will incline them to provide some information or do something unsafe.

As Jabir, Le and Nguyen (2025) show, the generation of phishing attacks has become more advanced because of the advances in generative artificial intelligence, making it possible that the difference of legitimate communications and malicious ones may become more challenging.

Sangwan (2024) stresses the importance of human factors in cybersecurity awareness and proposes in the opinion that cybersecurity awareness should be centered on the actual perceptions and responses of workers to cyber threats.

Insider Threats

Another significant human factor is insider threats since employees might have access to organisational systems and information through legitimate and lawful means. An insider threat

can be either deliberate or accidental and may occur by abusing access privileges, improper handling of information or neglecting to adhere to security practices.

Prabhu and Thompson (2022) state that insider threats constitute a distinct issue in cybersecurity because users that are legitimately permitted to access organisational resources already have privileged access to them.

The problem is also related to the organisational security culture. Georgiadesou, Mouzakitis and Askounis (2022) demonstrate that the culture of cybersecurity could assist in identifying the insider threats by providing an organisational framework in which the behaviours and risks associated with security could be assessed.

Behavioural Compliance

Employee compliance with organisational security policies also dictates the behaviour of employees as far as cybersecurity is concerned. Adherence can be in the form of password policies, access controls, information processing policies, incident reporting and other organisational policies.

Delso-Vicente *et al.* (2025) assert that there are several behavioural and technological determinants which dictate the adherence of the employees to the information-security policies as well as demonstrate that compliance is preconditioned by the combination of several conditions.

Cybersecurity compliance can also be influenced by the decision-making styles. Duzenci, Kitapci and Gok (2023) define the contribution of decision-making styles to the development of cybersecurity compliance behaviour, which means that employees can react differently to the cybersecurity demands based on the way they process and evaluate decisions.

Remote Working and Human Behaviour

Remote and hybrid work has been more problematic in terms of employee cybersecurity since the organisational operations are not taking place in the traditional controlled environments. Home networks, personal devices or other working environments can be used by workers to access organisational systems.

The cybersecurity awareness and training framework that Hijji and Alam (2022) have developed among the employees of remote working can be seen as the proof of the need to make the cybersecurity awareness strategies specific to the modified working conditions.

On the same note, Nwankpa and Datta (2023) target remote workers and find out the role of cyber awareness and cybersecurity policies in supporting safe behaviour in cases when employees are not working in regular organisational contexts.

The applicability of employee cyber-defence behaviour in working at home by Klein and Zwilling (2024) also supports the thesis that employee behaviour is the essence of cybersecurity in the face of technological controls.

2.4 Employee Awareness and Security Culture

Employee cybersecurity awareness is an idea that the employees are aware of cyber threat, security policies, safe practices and what to do in case of a possible incident. Consciousness is significant since the employees must have adequate knowledge to identify risks and know what is required of them.

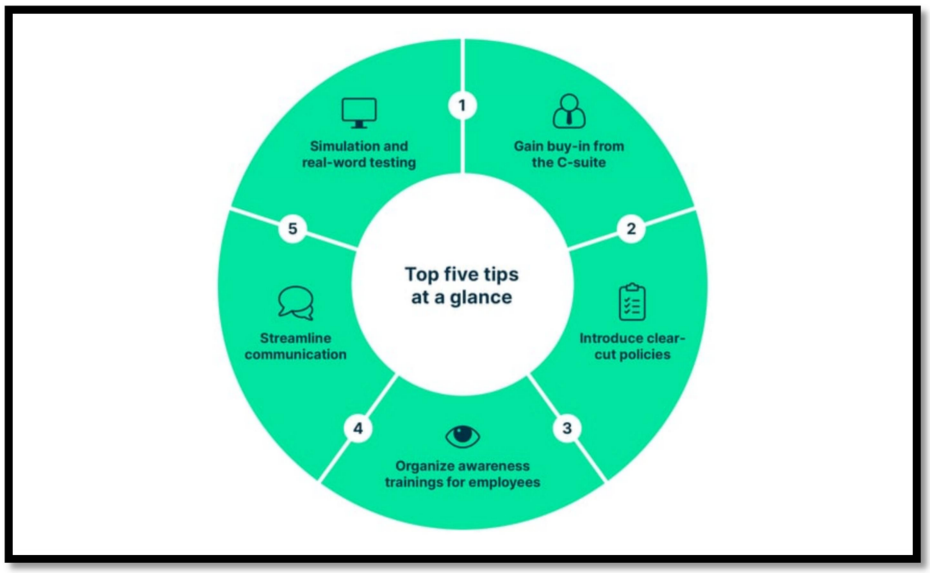


Figure 2.2: Employee Awareness and Security Culture

(Source: adnovum.com, 2026)

The importance of employee engagement and responsibility is another point that Abrahams *et al.* (2024) make about cybersecurity awareness and education programmes. Their work presupposes

that effective awareness should assume turning employees into active participants of the process of organisational cybersecurity rather than a passive receiver of information.

Employee engagement is thus closely related to awareness. Training may not create a tangible impact when employees do not consider cybersecurity as something that is relevant to their work. Tolossa (2023) focuses on the importance of cybersecurity awareness training among employees and training is one of the potential organisational interventions to reduce employee risks related to cybersecurity.

Awareness of employees, however, should not be regarded as a single training exercise. Taherdoost (2024) critically analyzes cybersecurity awareness models and training frameworks and proves the significance of systematic approaches towards cybersecurity education and awareness.

Of special significance is the difference between awareness and behaviour. It might be possible that the employees are knowledgeable about cybersecurity but fail to use them on a regular basis. Li, Xu and He (2022) discuss how antecedents and mediating factors influence cybersecurity protection behaviour and demonstrate that mediating factors do not have an influence on cybersecurity behaviour only because of basic awareness.

This relationship can also be explained by means of protection and motivation. Sulaiman *et al.* (2022) illustrate how Protection Motivation Theory is applicable to cybersecurity behaviour among government employees and the importance of protection motivation and responsibility in curbing cyberattacks.

Security culture makes individuals awareness at the individual level extend to organisational level. When cybersecurity is embedded in the organisational expectations, organisational values, organisational practices and in everyday behaviour, an organisational culture of cybersecurity emerges.

Aksoy (2024) assumes that cybersecurity culture has its merits in the development of resilient organisations that are not vulnerable to cyberattacks. This means that the awareness programmes should be undertaken in a larger organisational context that assists in fostering safe behaviour.

The management of an organisation is also important. Triplett (2022) attributes the importance of leadership to human factors in cybersecurity and states that the leadership of cybersecurity must consider the behaviour of the employees instead of focusing on technical controls.

Burton *et al.* (2023) also relate cybersecurity leadership and human factors and leadership qualities, implying that the leadership of organisations can impact cybersecurity performance by how they handle their employees, security expectations and organisational behaviour.

2.5 Cybersecurity Awareness Programmes

Cybersecurity awareness programmes should increase the awareness rate among employees in terms of cyber threat awareness and encourage more secure security behavior. These programmes may involve formal training, awareness, simulation attacks, educational materials, practical work and constant communication.

Hu, Hsu and Zhou (2022) provide a review of security education, training and awareness programmes and demonstrate the applicability of the knowledge of how cybersecurity education could influence employee security-related beliefs and behavioural intentions.

Training and Awareness

Ejiofor *et al.* (2025) highlight the significance of human factors in cybersecurity training and awareness and suggest that cybersecurity experts and workers need to have relevant knowledge and awareness to detect and act on cybersecurity threats.

On the same note, Rongala (2025) also outlines the best practices in cybersecurity training as well as the importance of employee awareness in reducing organisational cybersecurity threats.

Cyber Hygiene

Cyber hygiene is another means of spreading awareness among employees. It entails daily habitual activities that lessen vulnerability to cybersecurity threats, such as effective passwords, software upgrades, safe data disclosure and understanding of suspicious interactions.

Argyridou *et al.* (2023) create a cyber-hygiene framework of healthcare organisations and prove that the structured awareness strategies can facilitate cybersecurity and data-privacy awareness of personnel.

Kioskli *et al.* (2023) propose a human-centric perspective on cybersecurity hygiene in the healthcare context as well and underline the necessity to consider human behavioural factors to promote safe behaviour.

Training for Specific Organisational Contexts

Depending on the organisational circumstances, awareness programmes can be more effective. Small business and infrastructure employees or staff that handles critical infrastructure, telecommuters, and healthcare workers might have varied threats and responsibilities.

Waddell (2024) raises the question of cybersecurity education among healthcare workers and underlines the necessity to model education programmes in accordance with the individual human factors and security requirements of healthcare facilities.

Ugbebor *et al.* (2024) narrow down their research to small and medium-sized enterprises and suggest tailored employee training on cybersecurity awareness to minimise human error-related security incidents.

Measuring Awareness Effectiveness

The effectiveness of awareness should be measured to show whether the training has been effective. Organisations can tell how much the employees are knowledgeable, the security practices, the adherence to the policies, the reporting behaviour and other indicators.

Chaudhary, Gkioulos and Katsikas (2022) pay special attention to the development of measurement instruments to assess the cybersecurity awareness programmes and the importance of measuring the results of the programmes, rather than just training.

This assists in arguing out that awareness programmes should be evidence based and should be constantly reviewed. When an organisation cannot understand whether employees have enhanced their cybersecurity behaviour, it cannot easily find out whether the awareness programme is fulfilling its goal.

2.6 Theoretical Framework

2.6.1 Protection Motivation Theory

Protection Motivation Theory (PMT) is a worthy theoretical framework to the understanding of cybersecurity behaviour among employees. The theory is interested in the appraisal of threats

and protective responses by people before the decision is made on whether to act in a protective way or not.

Using Protection Motivation Theory to cybersecurity behaviour in government employees, Sulaiman *et al.* (2022) not only proves the importance of protection motivation and responsibility in understanding employee reactions to cyber threats but also illustrate the applicability of the theory in their research.

PMT is especially applicable to the field of employee awareness since awareness may also affect the way employees rate the severity of a cyber threat and how they rate the efficacy of protective measures. When an employee is aware of the repercussions of phishing and is convinced that reporting suspicious messages is a good defensive measure, then the employee might exhibit an increased likelihood of adopting this behaviour.

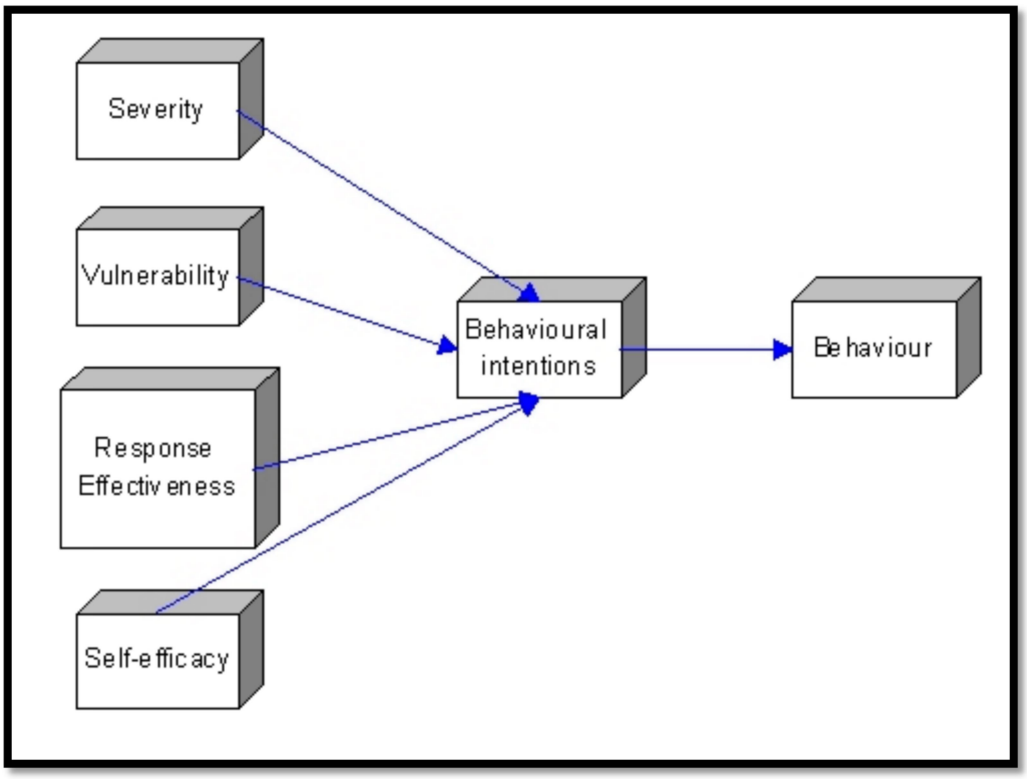


Figure 2.3: Protection Motivation Theory

(Source: garysturt.free-online.co.uk, 2026)

Li, Xu and He (2022) provide more reasons why the factors and mediating mechanisms influencing cybersecurity protection behaviour should be studied. Their results indicate that

cybersecurity behaviour is influenced by multiple antecedent and mediating factors as opposed to a single variable.

Employee awareness may influence threat understanding and attitudes towards protective behaviour and responsibility may influence whether it becomes secure behaviour.

2.6.2 Theory of Planned Behaviour

The Theory of Planned Behaviour is used to offer a complementary explanation of the employee cybersecurity behaviour. It assumes that behaviour is affected by attitudes, perceived behavioural control, and subjective norms.

Duzenci, Kitapci and Gok (2023) demonstrate that decision-making styles might influence cybersecurity compliance behaviour, which is applicable to the general argument according to which employee behaviour is conditioned by the way in which people make decisions on security issues.

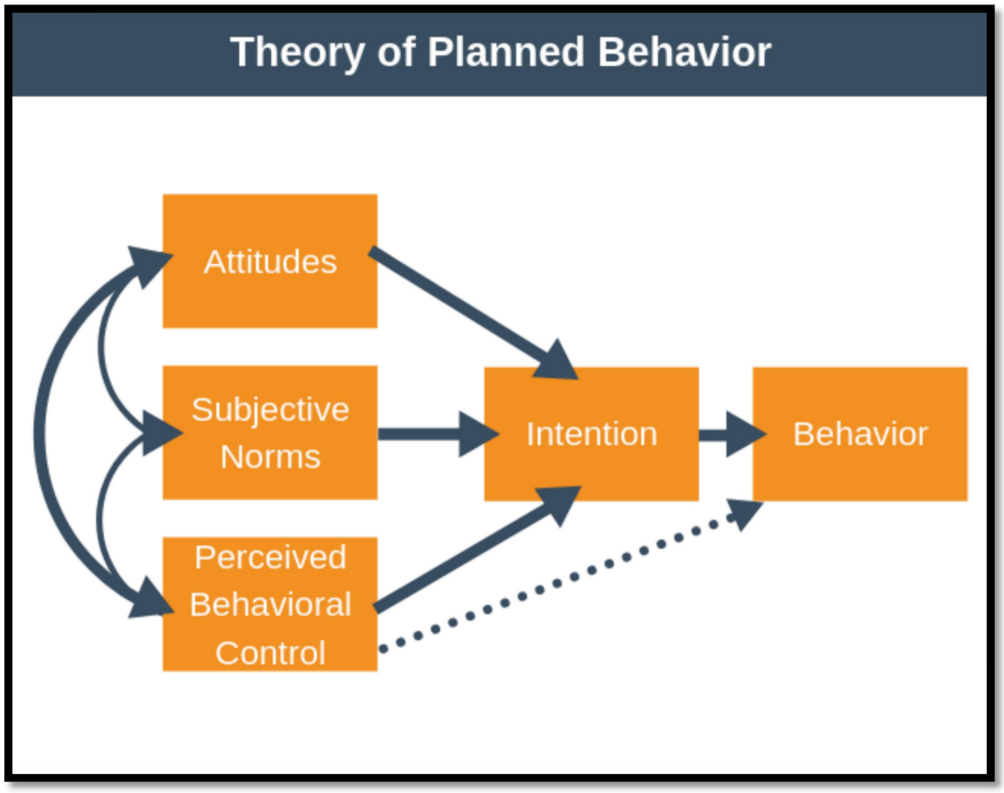


Figure 2.4: Theory of Planned Behaviour

(Source: expertprogrammanagement.com, 2026)

Yeng, Fauzi and Yang (2022) discuss the human element in cybersecurity compliance among healthcare personnel and illustrate that human factors are crucial in enhancing security practices in organisational settings.

TPB applies since employees are not alone when deciding regarding cybersecurity. The behaviour of the organisational expectations, peer behaviour, the management support and their perception of whether they can realistically comply with the security requirements can affect their behaviour.

The theory therefore makes PMT more insightful as it gives it the social and behavioural contextual factors of the cybersecurity practices of employees. PMT has a greater focus on the threat and coping perceptions than TPB that provides a model to comprehend the attitudes, social expectations and perceived control.

2.7 Literature Gap

The reviewed literature suggests that the human factors are of high interest, the employee consciousness and organisational cybersecurity, but there are areas that are not well integrated.

To begin with, the literature indicates that various human factors affect cybersecurity behaviour. Alsharida *et al.* (2023) name several points of view on human cybersecurity behaviour and prove that it is complicated to explain why certain security practices are embraced by employees.

Second, literature acknowledges the applicability of the cybersecurity awareness programmes, but the efficacy of the programmes will depend on whether the measurement of the level of awareness and whether the knowledge results in observable behaviour or not. Chaudhary, Gkioulos and Katsikas (2022) note the need to design suitable measures to assess awareness programmes.

Third, organisational culture is becoming a significant aspect of cybersecurity. Aksoy (2024) points out the significance of cybersecurity culture to organisational resilience, and Georgiadou, Mouzakitis and Askounis (2022) prove the importance of security culture to organisational preparedness and insider-threat detection.

Finally, the literature demonstrates that the awareness of employees cannot be viewed as such a feature that can be described as individual. The concept of human-centred approach to cyber risk

minimization supported by Peca and Tsurcanu (2025) justifies the need to focus on employees when considering organisational cybersecurity.

Synthesising the literature on three related dimensions, namely human error and vulnerability, effectiveness of the awareness-programme and the organisational security culture, this gap is bridged in this dissertation. This forms the basis of a study of how employee awareness is related to secure behaviour and diminished cyber threats.

2.8 Conceptual Framework

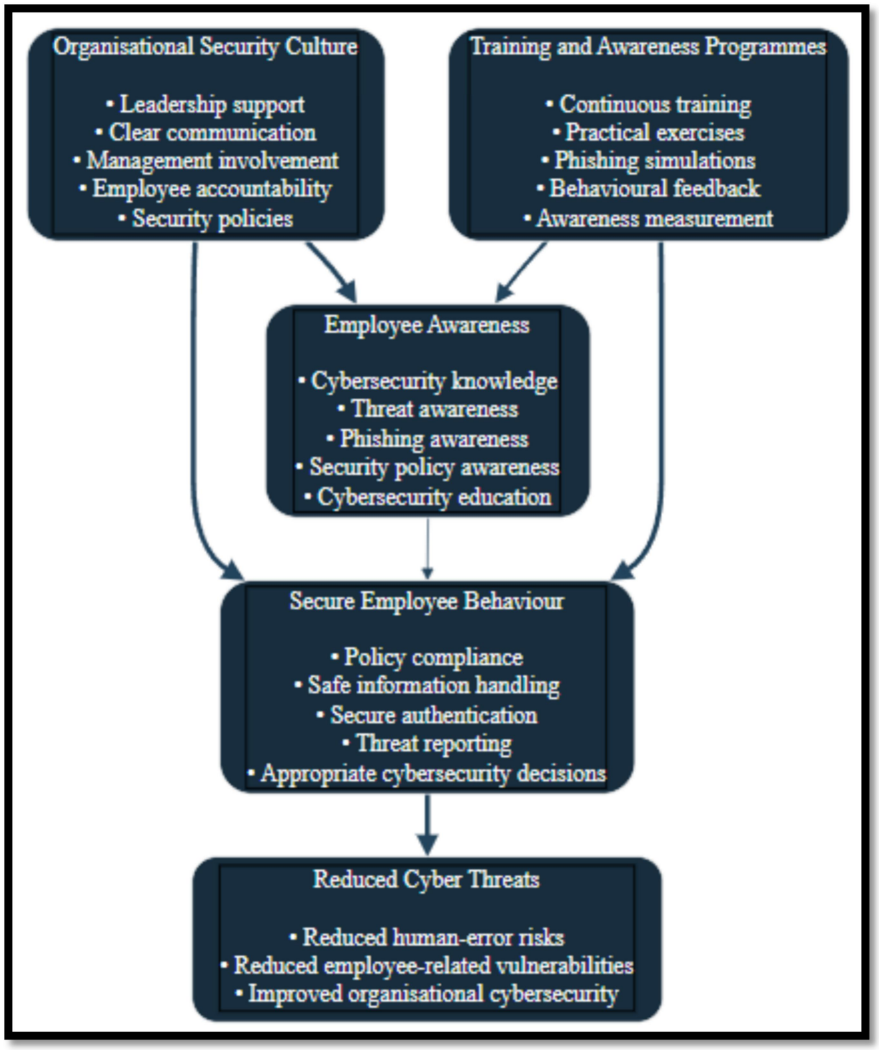


Figure 2.5: Conceptual Framework

(Source: Self-Created)

Chapter 3: Methodology

3.1 Introduction

The chapter describes the methods that will be utilised in the discussion of how awareness of employees can be applied to reduce cybersecurity risks in organisations. The study is based on existing academic literature rather than collecting primary data among staff members and companies in the specified project. It describes research philosophy, method, design, literature review plan, selection criteria and thematic analysis. It also outlines how Python-based software was used in literature validation, data cleaning, temporary key word coding, manual review and visualisation. The software is a research-support tool, but not an independent system to demonstrate cybersecurity assertions. The description of data quality, ethical and methodological limitations come at the end of the chapter.

3.2 Research Philosophy

The type of research philosophy used in the given work is interpretivist as it examines how awareness, behaviour and context of employees is handled in the existing literature concerning cybersecurity. Interpretivism would be suitable in the discovery of meanings and context rather than viewing cybersecurity behaviour as a technical issue. Human actions can be affected by the expectations at work, perceived risk, leadership, and support provided, and therefore, the results should be viewed in terms of their organisations (AL-Nuaimi, 2024). A review of opinions on human cybersecurity behaviour also aids in the analysis of personal and contextual factors (Alsharida et al., 2023). The philosophy thus leads the researcher to juxtapose the explanations and patterns between studies. The Python code will help to sort and code records, but the interpretation and validation will be left to the researcher.

3.3 Research Approach

An inductive approach is used in the process of identifying recurring ideas in the selected literature and building evidence-based themes. The research does not attempt to test a predetermined hypothesis, but rather investigates the reported findings on employee awareness, human error, and cybersecurity training, compliance, and organisational security culture. The approach allows similarities and differences of studies to inform analysis. The decision-making is considered a component of cybersecurity compliance in past studies (Duzenci et al., 2023).

Employee behaviour may also be impacted by stress, burnout, and security fatigue (Nobles, 2022). This analysis is then transferred to the results to overall themes that can give responses to the research questions. The software aids preliminary key word suggestions, which are tentative and need to be cross-examined with the source evidence prior to a theme being considered as validated.

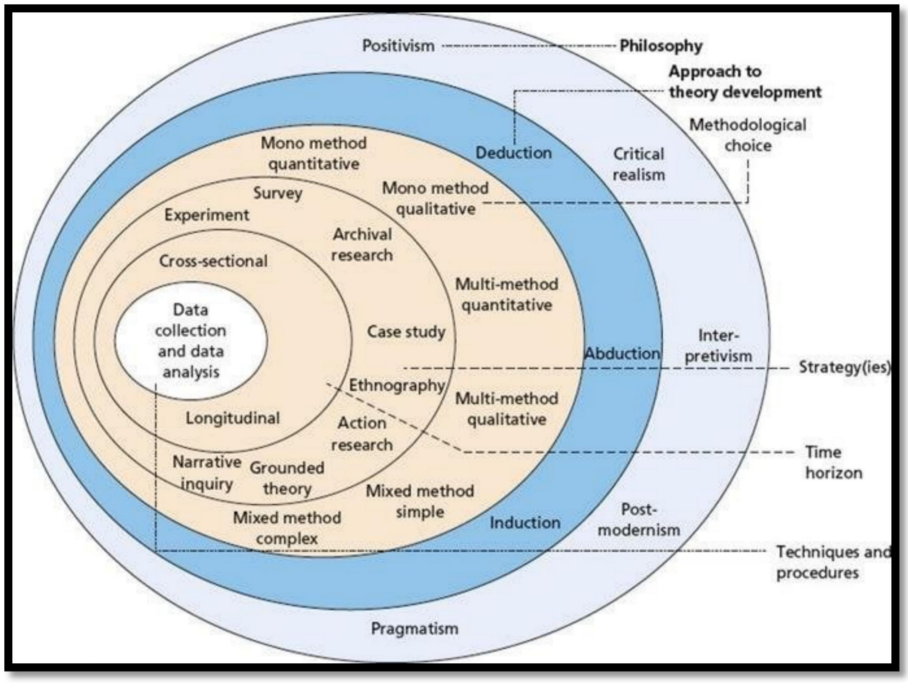


Figure 3.1: Research Approach

(Source: researchgate.net, 2026)

3.4 Research Design

The research uses a qualitative secondary design based on a systematic literature review. It looks at published research, rather than collecting surveys, interviews and experimental data regarding employees. The research design is appropriate because the study aims to elaborate on the available data on the relationship between employee awareness and organisational cybersecurity. A systematic review process helps to make the literature search, screening and selection more transparent. The study is focused on academic publications in English that were written mainly 2022-2026 and that are related to the human factors, employee awareness, training, cybersecurity behaviour, or security culture. The selected literature is compared with the help of thematic analysis. Python scripts are used to assist in the validation, cleaning and coding candidates,

preparing the data to be examined by humans and producing charts. The quality, relevance and breadth of the sources used, therefore, limits the conclusions of the design.

3.5 Research Strategy: Systematic Literature Review

This research uses a systematised literature review to determine and analyse published work on employee awareness and organisational cybersecurity. The plan will provide a methodical approach of searching and identifying, selecting and tabulating relevant scholarly materials. Scopus, IEEE Xplore, Springer, ScienceDirect and Google Scholar are covered in the literature search. The keywords will be cybersecurity awareness, human factors, behaviour of employees, security training, policy compliance and organisational security culture. Pre-determined inclusion and exclusion criteria are used to select relevant publications. Duplicates of records are identified and removed, and titles/abstracts and full texts are compared with eligibility requirements. Transparency and traceability in the selection process is improved by the PRISMA principles. The Python program can be used to validate literature and clean data, group data by key words, review and write research summaries and charts manually.

3.6 Inclusion and Exclusion Criteria

The inclusion and exclusion criteria were set to ensure the relevance of the literature that has been chosen. Publications included are about employee awareness, human factors, cybersecurity behaviour, training, compliance, or organisational security culture. The preference is given to peer-reviewed academic articles in the English language published in 2022-2026. Any source must be valuable to at least one of the research objectives. Publications will be excluded when it is outside of the selected period but there is no explicit theoretical or methodological rationale, it is not in English or is entirely technical cybersecurity that lacks much human aspect. Articles, which are not related to organisational or employee cybersecurity are duplicated, and those where there is lack of scholarly information are also removed. These standards facilitate the standard literature screening and only those sources are included which will give plausibility to the anticipations of the researcher in his or her findings.

3.7 Literature Search and Selection Process

The search engines used in the literature search are Scopus, IEEE Xplore, Springer, Science Direct and Google Scholar that offer publications on employee cybersecurity awareness. Some

of the key words are human factors, phishing, security training, employee behaviour, policy compliance, and organisational security culture. Records that have been retrieved are sorted out prior to duplicating identities and missing information. The abstracts and titles are served on the set inclusion and exclusion criteria. Evaluation of relevance of relevant publications is then done through review of full text to determine relevance to the research. The PRISMA principles will be used to report on the selection process and make it transparent and consistent. Python programs assist in validating literature fields, checking duplicate papers, cleaning of research records and preparing output.

3.8 Data Processing and Keyword-Based Classification

A Python-based workflow is used to process the data in the research to organize literature records and allow the use of the same classification. Data cleaning and validation (checks of missing data, duplicate paper identifiers and validity with respect to year of publication) start the process. The program subsequently uses key words matching the chosen research areas to come up with tentative candidate classifications. These recommendations can assist in the organisation of records based on pertinent notions, such as human error, awareness programmes, and organisational security culture. The create_review_file.py program converts the records to be scored by hand, in which the researcher may enter the confirmed classifications, evidence to support the classification, whether the review is approved or not, and comments. The eligible reviewed records are identified with the help of the next scripts, and the summaries of the tables and visualisations are generated. The matches of keywords are preliminary as they are based on a predefined set of keywords and do not consider the contextual meanings. The final products help in formal reporting but are not causally related nor effective in proving the effectiveness of cybersecurity measures.

3.9 Software Design and Implementation

3.9.1 Software Architecture and Project Structure

The Python script was used as the data analysis tool to help in the systematic organisation, processing, validation and review of cybersecurity literature. The workflow consists of seven scripts: main.py, validate_literature.py, data_cleaning.py, thematic_analysis.py, create_review_file.py, verified_summary.py, and visualise_results.py. The thematic analysis is a

temporary classification in terms of keywords, but not a thematic analysis, although the script is titled thematic_analysis.py. The scripts create formatted CSV and graphical data to aid research documentation and interpretation.

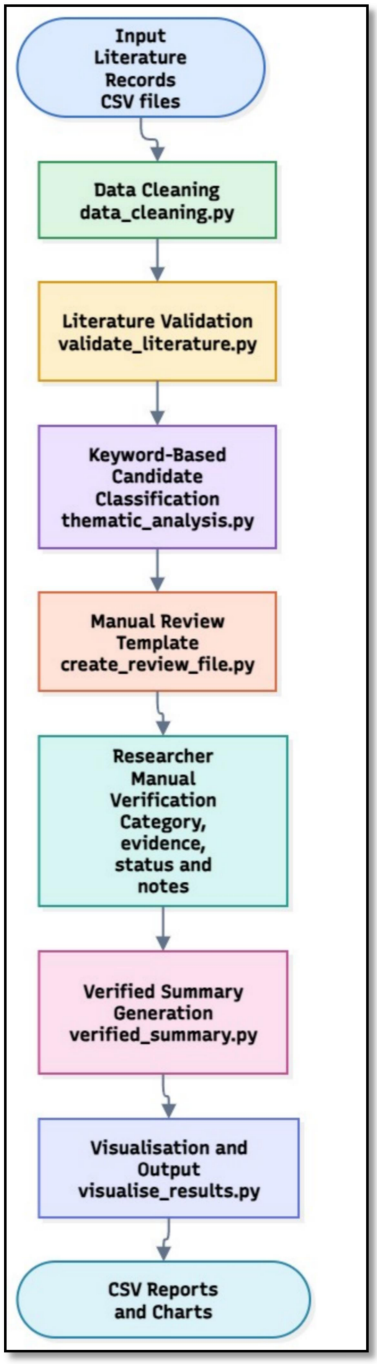


Figure 3.2: Python-Based Research Software Workflow

(Source: Self-created)

3.9.2 Data Cleaning and Literature Validation

The process of data-cleaning will make the text fields uniform, remove duplication of records and identify information that is vital and missing. These functions improve reliability and facilitate planning of literature records to be processed in the future. The validation script validates the record identifiers, year of publication, required fields and potential duplication. However, automated validation cannot be the sole credible and relevant factor of not all publications.

3.9.3 Keyword-Based Coding and Manual Verification

The software applies pre-determined criteria of matches of the keywords to identify the candidate categories in the literature records. This operation helps to pre-organise data related to employee awareness, human error and organisational security culture. The matches on keywords are, however, tentative since a match of a keyword does not necessarily imply that a publication promotes a specific interpretation. The software thus creates a manual review template where the researcher can note a confirmed category, backup evidence, review status, and notes. The presence of human verification is needed to determine whether the extracted evidence is really going to support the suggested classification. Only those records that satisfy the verification requirements that are imposed should be included in verified summaries.

3.9.4 Verified Analysis, Visualisation and Output Generation

The last processing phase produces records that meet the appropriate eligibility criteria into summaries and charts. The verified_summary.py script summarizes the verified classifications and the visualise results.py creates the graphical representation of the results available. CSV output can also be used to further explore and obtain traceable records of the process. The current outputs are eight clean records and eight candidate classifications, but the verified output has two records with only one of them being eligible to be verified.

3.10 Data Extraction Framework

Data-extraction framework was used to categorise the information in the literature of choice in a systematic fashion. The structure captures bibliographic identifiers and other publication information and data needed to classify and review. The extracted information is transformed into structured CSV files, which allow verifying records, comparing and tracking them in the

working process. The verified category, supporting evidence, review status and researcher notes are in the template of the manual review. This approach separates the data mixed out of the publications and the provisional groupings that have been made because of matching by key words. It also differentiates between records that must be reviewed and those that can be verified in summaries. A fixed extraction framework enables systematic records and eliminates possibilities of different records management.

3.11 Trustworthiness and Quality Assurance

Formal validation of data, policy of uniform data processing, history of review domains, and separation of provisional and approved classifications helped in facilitating credibility. A field-check validation and checking against duplicates helped to detect potential problems with data-quality before analysis. Manual verification was an additional check whereby supporting evidence and a completed status of review must be provided prior to qualified records being added to verified summaries. The processes make the process more transparent and allow verifying outputs with their records. Automated matches of keys can, though, overlook pertinent evidence or match words out of context. Quality assurance should therefore be approached with care and on a case-to-case basis and incomplete verification reported.

3.12 Ethical Considerations

The research is based on secondary literature and does not include the actual recruitment of human subjects. Proper reporting of published results, proper recognition of authors and not falsification or false heedfulness of evidence are all part of ethical practice. The manual verification is carried out to increase transparency by recording the supporting evidence. Research records are kept in a systematic way and only based on the evidence reviewed can conclusions be made.

3.13 Methodological and Software Limitations

The limitation of the research is due to the quality and the breadth of literature selected and records retrieved. The classification based on keywords can miss context or label pertinent publications. Manual verification is not done and there is only a single record that can be verified and analyzed. Output inconsistencies can also be due to variations in script inputs and eligibility rules. Therefore, interpretation of results should be done with caution.

3.14 Chapter Summary

This chapter outlined the research design, selection of literature, data extraction and Python-based processing workflow. It distinguished between the provisional classification of keywords and the manual verification and the quality assurance, the ethical aspects as well as the methodological constraints. Chapter two gives us the results we have with a point of view towards the limited evidence that has been verified.

Chapter 4: Analysis and Findings

4.1 Introduction

The chapter includes the results of the Python-based program created to facilitate the study of the employee cybersecurity awareness and cyber risks related to human factors. It examines the literature register, the research data, search plan and verified document of paper review. The data is analysed in the light of the data organisation, literatures selection, classification of the candidates, verification of a person and consistency of the existing files. Human and contextual factors are noteworthy when it comes to cybersecurity in organisations (AL-Nuaimi, 2024). In this respect, the chapter draws a line between the information recorded in the software and to conclusions drawn out of analyzed literature.

4.2 Research Dataset and Output Overview

Four files were analyzed to get a glimpse into the process of research, which was supported by the software. The literature register table (literature_register.csv file) will be a representation of two publications, P001 and P002, and nine variables that will contain bibliographic information, year of publication, database, screening decision and exclusion reason. Both publications have an Include decision. The research_data.csv has eight entries (P001-P008) in five columns; paper identifier, year, topic, findings and type of source.

The search_strategy.txt file contains search questions, five different combinations of search-terms, proposed search-academic databases, and inclusion/exclusion criteria. These criteria dwell on the relevance to employee cybersecurity awareness or human factors, appropriate publication information and evidence available. The verified_paper_review.csv file contains two records with ten records that include the category of candidates, key words matched, verified category, supporting evidence, review status and reviewer notes.

These files have various functions. The literature register is where the screening decisions are registered and the research dataset is where brief summaries of eight records are presented. Assessment information is further documented in the verified review file and is in two publications. However, the difference in the years of publications and issues in files indicates that the files are not perfectly matched.

4.3 Literature Validation and Data Cleaning Results

There are two records in literature register all of them Include. The article by Hillman, Harel and Toch, evaluating organizational phishing awareness training on an enterprise scale, was published in 2023 in Computers and Security and it is defined as P001. The 2025 conference paper of Ho and colleagues, Understanding the Efficacy of Phishing Training in Practice, in the 2025 IEEE Symposium on Security and Privacy is P002. All these documents have a DOI link, and the screening database is Google Scholar accredited.

Exclusion-reason field in P001 is filled in as “not applicable” whereas in P002 it is left blank. The lack of the records does not necessarily indicate that the screening decision was incorrect since both records exist. But the regular field control would enhance visibility of the register.

The research data consists of eight phishing records, awareness training records, security culture records, password security records and social engineering records. All the eight records contain the source type of Journal. But discrepancies of the files available must be resolved. The date of P001 in the research_data.csv is 2024 and the register and verified review file indicates that the Hillman study is 2023. The research dataset has P002 dated 2023 whereas in the other two files, the date is 2025. The phishing training effectiveness is also different from the Awareness training in terms of P002 topic.

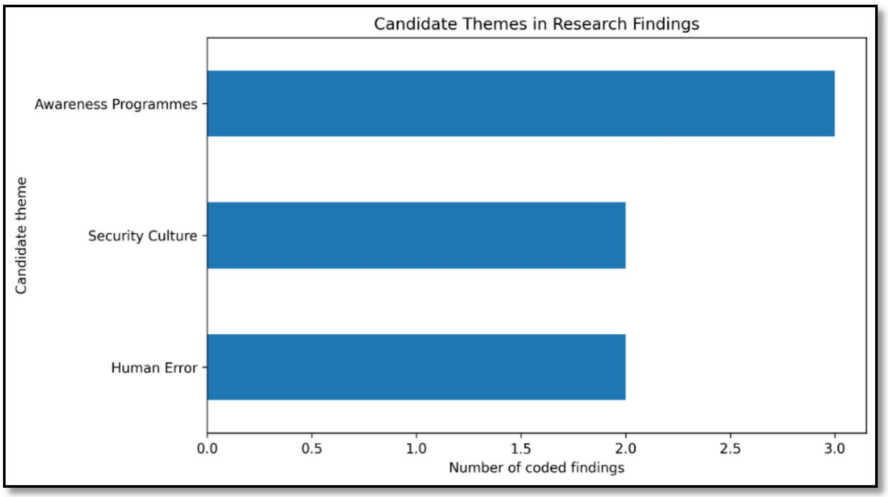


Figure 4.1: Candidate Classification Results

(Source: Vs-code)

Such discrepancies could be different records, provisional summaries or errors during data entry; the files at hand do not tell how. Consequently, the records need to be compared with the original publications prior to the final analysis. Bibliographic data plays a crucial role in the analysis of human and contextual factors of cybersecurity (Alsharida et al., 2023). The current files suggest quite incomplete literature organisation and screening but do not show that all the eight research records have been completely validated.

4.4 Candidate Classification Results

Two publications have the types of candidates and similar keywords, which are recorded in the reviewed document. P001 has the candidate categories “Awareness Programmes” and “Phishing Susceptibility”. Its key words synonyms are phishing awareness training, personalised phishing emails and click-through rate. These abbreviations indicate that the record is on the response of the employees to the simulated phishing messages and evaluation of awareness training.

P002 is a combination of three categories of applicants called; Awareness Programmes, Training Effectiveness and Phishing Susceptibility. The following are the matched keywords that are logged: awareness training, phishing training and training effectiveness. These categories presuppose that they could be applicable to the research question, but do not solely define the results and effectiveness of the training intervention.

The categories of candidate and verified differ. The key-word classification would help in classifying publications using the help of the relevant terms but then the manual review is needed to identify the availability of evidence to back the proposed classification in the publications. The behaviour of human cybersecurity is determined by various personal and contextual factors, and therefore, isolated keywords might not be sufficient to reflect the situation in a study (Alsharida et al., 2023).

The approved review file that is in place at the time documents the two records of candidates as opposed to eight records. There are eight records in the separate research dataset, which are not all provided in the uploaded verified review file with candidate classifications. The current evidence will not allow concluding about a predictable splitting of the classifications of the whole dataset. Types of applicants ought to be considered as an initial organising system in comparison to confirmed results of employee consciousness or cybersecurity danger.

4.5 Manual Review and Verification Results

The results of the manual review indicate different levels of accomplishment of the two books. P001 is labeled Reviewed and the verified label Phishing Susceptibility. Its evidence field mentions that the abstract and discussion provided in the article have higher click rates of personalised simulated phishing emails in comparison with general emails. The notes were recorded that personalised phrasing was associated with a 1.5-2 times increased number of clicks and full-text review before final dissertation coding (Hillman, Harel and Toch, 2023).

This result provides a written basis to the rough classification, but the examination documents limit the strength of the evidence as the whole article is not under examination. The discovery should then be considered as a tentative evidence-based classification but not the ultimate assessment of the entire research.

P002 remains Pending. There are categories and keywords of its candidates which are stored but the verified category and supporting evidence fields are empty. The clearly mentioned reasons provided by the reviewer mean that an entire paper needs to be checked and evidence recorded before the final classification can be assigned to it.

Consequently, the file contains two reviewed records in the sense that each of them contains a review record, yet only one with a verified classification done; P002 is not a complete record. This is a crucial difference in reporting the progress of the software. The workflow facilitates segregation of candidate identification and documented verification, and further investigation is needed before either of the publications can become part of approved summaries. The results also support the rationale to assess awareness training based on the relevant evidence instead of believing that taking part in it automatically decreases phishing vulnerability (Abrahams et al., 2024).

4.6 Verified Classification and Summary Results

The qualified review file located has found two publications but with different review status. P001, Evaluation of Phishing Training at an Enterprise Level, is assigned the type of verified Phishing Susceptibility. P002, The Effectiveness of Phishing Training in Practice is waiting as the category of the proven and its evidence remains uncompleted.

The evidence captured in the case of P001 is that employees were more likely to often click a simulated phishing email with personalised wording than they were to random emails. The review indicates that the abstract and discussion provided by the publisher reported approximately 1.5 to 2 times greater number of clicks of personalised email. These findings are in the review file about (Hillman, Harel and Toch, 2023).

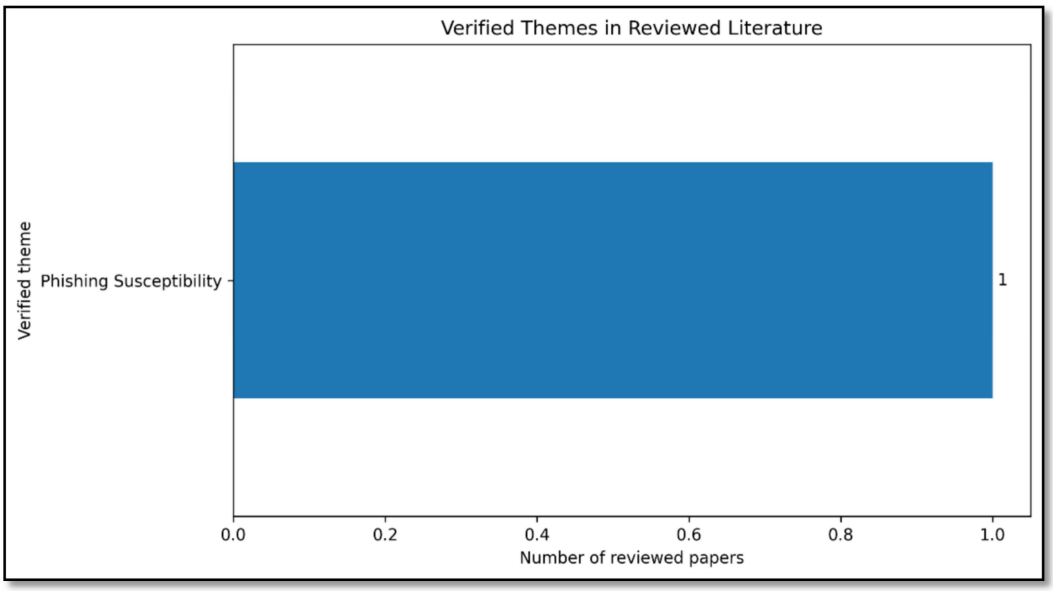


Figure 4.2: Verified Classification Results

(Source: Vs-Code)

The P002 record describes a randomised controlled trial including ten simulated phishing campaigns and eight months of randomisation with more than 19, 500 employees in a large healthcare organisation. But the proved type and evidence sphere are vacant. Therefore, these facts should be viewed as the data, which is included in the dataset, but not the results of the review, which are independently checked. The journal will be accepted as a conference paper in 2025 (Ho et al., 2025).

The overview with the existing data backing it is thus constrained. One of them has been reported to have a valid classification and the other is under consideration. There are eight rows of research data csv that cannot be filled in all the verified summaries as only two are found in the verified review file. In addition, P001 and P002 do not match in the years of publication, or description of the topic, in files.

4.7 Visualisation and Dashboard Output Analysis

The given CSV files assist in creating a visual perspective on the candidate types and tracing the development. In the verified review file P001 there are two candidate categories Awareness Programmes and Phishing Susceptibility. P002 entails three: Awareness Programmes, Training Effectiveness, and Phishing Susceptibility. When the counts of categories of candidates are on a one-on-one basis, there will be both Awareness Programmes counted and Phishing Susceptibility counted and Training Effectiveness counted.

These counts are candidate classifications of the two records that are reviewed and not the distribution of categories in all the eight records of the research dataset. Phishing Susceptibility is verified in the category field of P001 but P002 is yet to be classified. These records would therefore result in a single Reviewed record and a single Pending record on a review-status chart.

They can be employed to identify the unfinished work in review and match the candidate's classification to the confirmed results. The given CSV files, however, do not assure the final appearance or accuracy of a dashboard or a graphical output as such. The chart needs to be matched with the underlying data particularly when the names of the categories are contained in the same fields using semicolons. The visual display can be used to complement inspection and cannot replace the evidence checks or the inconsistencies in the traces of origin.

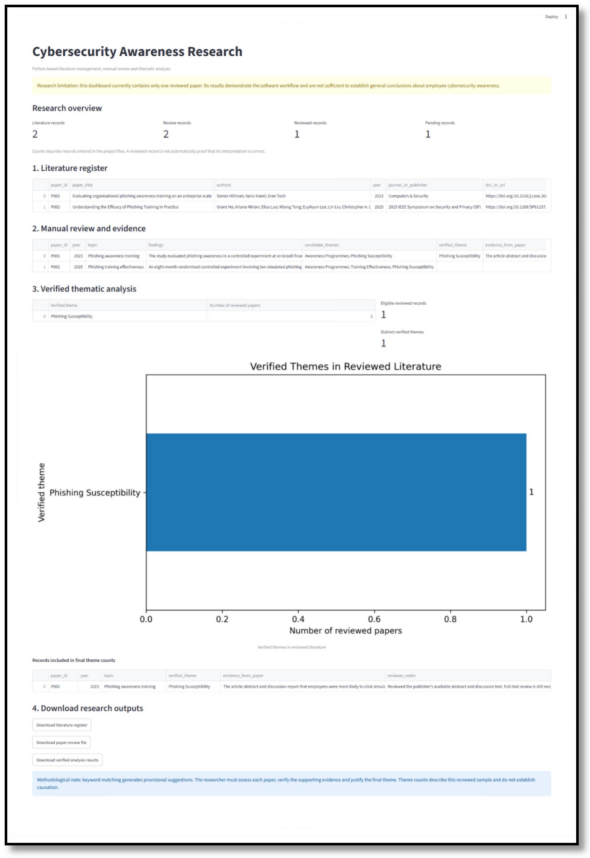


Figure 4.3: Cybersecurity Awareness Research Dashboard

(Source: Vs-Code)

4.8 Analysis of the Three Predefined Research Categories

The research framework reveals three predetermined categories categorizing literature in the field of employee cybersecurity awareness as human error and cybersecurity risk, employee awareness programme, and organisational security culture. These groups impact literature analysis and the organisation of groups of candidates. These are not created in a formal thematic analysis. The available records of phishing awareness training provide more first-hand data than any other area in the three categories and therefore there must be a distinction between the data on records and the rest of the literature.

4.8.1 Category 1: Human Error and Cybersecurity Risk

Human factor and behaviour of staff members are also materials of concern when it comes to cybersecurity since humans are the ones who have direct interactions with organisational systems,

safety surveillance and even malicious messages. Among the research databases is a P001 summary by which organisations may be susceptible to phishing attacks due to human error. It has another indicator of P007 summary which is that social engineering is capable to manipulate employees to give sensitive information.

The human factors research point to the importance of considering the behaviour of employees when dealing with cybersecurity (Edeh, 2023). The decision-making styles can also influence cybersecurity compliance behaviour (Duzenci, Kitapci and Gok, 2023). These sources give a literature-grounded justification for keeping human error as a pre-determined category.

The confirmed review file involves more detailed evidence on phishing vulnerability. P001 records the number of clicks on personalised simulated phishing messages by the employees and provides the category Phishing Susceptibility (Hillman, Harel and Toch, 2023). The review of the sources is not complete, and the evidence cannot be applied to all cyber threats that are related to humans.

The Python workflow could help to locate and filter the relevant records, but the classification of candidates is not a validated decision. P007 and other relevant publications will have to be further validated in the study to be able to conclude in general on human error, phishing vulnerability, or social engineering.

4.8.2 Category 2: Employee Awareness Programmes

Employee awareness programmes are meant to make sure that employees are not ignorant about cyber threats, and they acquire correct security behaviours. Three items in the research data.csv, P002, P005 and P008 are defined as Awareness Training. Their summaries respectively state the significance of identifying suspicious email, practice understanding the threats, and the behavioural measures of evaluating the training. These are short descriptions of datasets, not the results of research which was independently validated.

P001 is checked review file, and it refers to the organisational phishing awareness training. The fact that it has documented shows that personalised messages with phishing were more likely to be clicked as compared to the general messages, which implies that reactions of employees may differ based on the type of simulated attacks (Hillman, Harel and Toch, 2023). This observation can be extended to the assessment of the awareness-raising programmes whereby the assessment

of training should be on the behaviour of the employees but not necessarily on the attendance of the programmes.

The works of the employee engagement and accountability in cybersecurity awareness and education programmes are also presented in the reference list (Abrahams et al., 2024). This literature aids research on the design and evaluation of awareness programs even though not all the summaries in research dataset are validated. The nature of candidates makes it easy to determine the publications that can be further reviewed like P002 awaiting.

4.8.3 Category 3: Organisational Security Culture

Organisational security culture is associated with expectations, policies, leadership practices and shared responsibilities that affect the cybersecurity behaviour of the employees. P003 which summary is that leadership support can provide an impetus to secure behaviour among employees and P006 which is that clear policies can enable adherence to cybersecurity are reflected in the research data. These entries present what the category should cover yet they have not been published in the given research dataset.

The organisational cybersecurity culture studies highlight the applicability of the culture to organisational resiliency to cyberattacks (Aksoy, 2024). The human and contextual factors also require inclusion of cybersecurity in organisations (AL-Nuaimi, 2024). They are the sources that enable maintaining organisational security culture in the form of a research category.

The current verified review file, which is in its verified state, does not contain a verified classification of organisational security culture. Its two candidate review records are interested in awareness programmes, training effectiveness, and susceptibility to phishing. As a result, the information provided about software is unable to reach an accepted decision on the influence of leadership, organisational policies, or security culture.

Even now the software can help this part of the study by helping with the organisation of the records that will apply in the study, and which pieces of information should be checked. Nevertheless, P003 and P006 should relate to verifiable publications and cross-examined with their primary evidence to be able to make more solid conclusions. Organisational security culture should be met with the help of the literature discussed to the last drop and should be considered as a sphere that should be verified at the level of records further.

4.9 Evaluation of the Software Workflow

The Python-based workflow offers a systematic method of structuring cybersecurity literature, documenting screening decisions, classifying candidates, and reporting about manual review. Two records were registered to be included, and eight records summarised in the research database. The authenticated review record offers more evidence fields and review status of two publications.

Separating the candidate classifications and verified classifications is a strength. P001 is supported and reviewed, P002 is Pending. This distinction reduces the chances of listing all the keywords match as valid results. Cybersecurity risk assessment is also something to take into consideration the behaviour of the employees and the context of organisations (AL-Nuaimi, 2024).

The workflow is restricted though. There are differences in the year of publication and describing of the topic of files, and the bibliographic information of P003-P008 is not provided in the research dataset. The notes of P001 indicate that a reader should read the text in its entirety. These issues limit the power of conclusions and traceability. The workflow should subsequently contain standard identifiers, resolved metadata, complete source citation, and documented evidence, to allow larger research inferences to be made.

4.10 Findings in Relation to Research Objectives

The existing outputs facilitate the aim of structuring the pertinent cybersecurity literature and recording its screening and review. Eight succinct records on the same are available in the research data set and two publications were included in the literature register. The areas that may be relevant will be identified by candidate classifications and the supporting evidence of P001 and leave P002 will be recorded in the approved file of review as Pending.

The outputs thus show the progress of systematic review process but fail to ensure full validation of all the eight records. Lack of bibliographic information and discrepancies in files should be checked further. Human cybersecurity behaviour research warrants considering personal and situational conditions in the explanation of security employee practices (Alsharida et al., 2023). Generalisation of knowledge regarding awareness programmes, human error or organisational security culture needs further verification of sources.

4.11 Chapter Summary

This chapter examined the research information, literature register, search strategy and verified review file. The outputs reflect the organization of records, classification of candidates and the recording of decision of the review in Python workflow. P001, however, has been successfully tested and P002 is awaited. The time lapses between the year of publication and the descriptions of the topics and lack of source information in other records limit the conclusions that can be drawn. The next chapter must critically read the evidence presented and make a distinction between evidence that can be established by the records and overall conclusions by the literature.

Chapter 5: Discussion

5.1 Introduction

The chapter describes the relevance of the research results to the awareness of employees in relation to cybersecurity and the ways in which the cyber risks can be reduced depending on human factors. It links the outputs of the available software to pertinent academic literature and research questions. The discussion considers employee awareness, human factors, awareness programmes and potential improvement measures. Cybersecurity in organisations is a topic that is significant in human and contextual factors (AL-Nuaimi, 2024). The chapter also talks about Protection Motivation Theory and Theory of Planned Behaviour that may be employed to explain the behaviour of employees regarding security. The shortcomings of the existing software outputs are admitted, especially the unfinished checking of research records and discrepancies of the existing data files.

5.2 Discussion of Research Question 1: Employee Awareness and Organisational Cybersecurity

Awareness of cybersecurity at the employee level can also contribute to organisational security since it will allow employees to detect the threats, understand the security procedures, and make better decisions when handling digital information. Awareness is particularly relevant when the employees encounter phishing messages, suspicious links or information requests. However, the level of awareness on security risk does not necessarily mean that employees always will act accordingly.

The research on human cybersecurity behaviour indicates that individual and environmental factors influence the decisions that employees make regarding the issue of security (Alsharida et al., 2023). This has been a pointer that understanding awareness has to be considered within the circumstances of the workplace, the organisational expectations and the opportunities that the employees possess to make use of their knowledge. Employee engagement and accountability is another aspect that should be considered when analysing the cybersecurity awareness and education programmes (Abrahams et al., 2024).

The existing research documents give a small sample of this relation. The validated test of P001 is the data concerning the response of the staff to individualised simulated phishing email. The

review indicates the abstract and discussion they provide indicate that personalised messages are more likely to get clicks compared to general messages, and that full-text review should be recommended (Hillman, Harel and Toch, 2023). This fact does not establish the overall effect of awareness training on cybersecurity in an organisation but suggests that employee responses may vary based on the characteristics of a potential threat.

The program helps to enlist potentially valuable publications and supporting evidence documents to peruse. Nevertheless, only one record has a confirmed classification as of now and P002 is still to be classified. The software output available cannot prove that increased employee awareness has decreased cyber incidents in organisations. The overall literature encourages researching awareness as a part of management of cybersecurity, organisational circumstances, and adequate protective technologies (Aksoy, 2023).

5.3 Discussion of Research Question 2: Human Factors Behind Cybersecurity Threats

The interaction between security technologies, organisational policies and digital communications is dependent on human factors. The concerns are decision-making, attention, security knowledge, compliance and responses to potentially malicious messages. The failure to detect red flags or follow best security practices can contribute unconsciously by workers to the unintentionally introduced vulnerability.

Human factors are identified as a considerable factor in cybersecurity management in organisations too (Edeh, 2023). The decision-making styles can also influence the compliance of employees with cybersecurity, meaning that individuals might act differently towards security requirements (Duzenci, Kitapci and Gok, 2023). These results foster an approach that considers the organisational environment and behaviour of the employees.

The current research evidence comprises of the logs of phishing, social engineering, password security and culture of security. These entries lead to the fields of investigation in question, though their summaries cannot be considered the independently verified findings until their bibliographic information and the evidence can be confirmed. The confirmed review file provides a more particular example with P001 that falls under Phishing Susceptibility. The

review notes however indicate that the classification should be finalised after full-text analysis is conducted (Hillman, Harel and Toch, 2023).

The Python workflow helps to handle records with categorisation of candidates and provides evidence-recording and review-decision fields. Nevertheless, a search term that is similar cannot be used solely to tell why workers behave in a particular way or which human factors influence cyber-incidents the most. The minimal amount of evidence that has been validated also cannot be used to make a good comparison between phishing, social engineering, password practices as well as organisational culture.

5.4 Discussion of Research Question 3: Effectiveness of Awareness Programmes

Employee awareness programmes on cybersecurity have the potential to promote awareness of cyber threats among employees, although their effectiveness must be measured through the behaviour of employees. Knowledge about phishing, suspicious links, password safety, and reporting habits is valuable, but the knowledge that is acquired during training might not necessarily translate into consistent safe behavior. Hu, Hsu and Zhou (2022) emphasise the applicability of security education and awareness programmes to the security-related beliefs and behavioural intention of the employees.

Practical training and more opportunities to train security knowledge can help the employees to respond to threats more appropriately. Cybersecurity training has been associated with the implementation of security practices, yet the results have to be communicated in the context of the approaches and results analyzed in independent research (Peca et al., 2025). The effectiveness of the training should hence be measured on the foundation of threat recognition, reporting suspicious messages, adherence to and behaviour change against course completion. Chaudhary, Gkioulos and Katsikas (2022) stress that it is essential to design metrics to evaluate cybersecurity awareness programmes.

The software records posted do not provide much evidence to base the effectiveness on. P001 is concerned with the organisational phishing training and records an outcome in reference to the rate of increase in the number of clicks in simulated personalised phishing email, in comparison to general messages. However, it is mentioned in the review that it is better to read the whole text and then decide on the classification (Hillman, Harel and Toch, 2023). P002 concerns the

effectiveness of phishing training, but it remains not yet confirmed, and the category and evidence to prove it are not finalized.

The python workflow helps to systematize the pertinent publications and write down evidence that will be reviewed but does not experimentally test training interventions. In this way, the awareness programmes should be addressed in terms of their effectiveness primarily on the basis of the reviewed literature. Taherdoost (2024) emphasizes the importance of systematic methods of cybersecurity awareness and training.

5.5 Discussion of Research Question 4: Measures to Improve Employee Awareness

A mix of education, practice, organisational support, and continuous assessment is necessary to enhance employee awareness of cybersecurity. An increase in frequency of training can help the employees to detect emerging threats, but job specific information can be employed to act on the risks in the various jobs. There might be employees that deal with sensitive information, or are exposed to frequent outside communication, and as such, they may need training relating to those tasks. Ejiofor et al. (2025) highlights the role of training and awareness in dealing with human factors in cybersecurity.

They can offer practical exercises to the employees to enable them to put to use whatever they have learned when they encounter suspicious emails and other threats. Phishing activities can also be hypothetically carried out to identify the areas that need additional instruction. The training should be supported with clear procedures on what to do when suspecting an activity so that the employees can know what to do in case of doubt. Rongala (2025) explains the training practices related to cybersecurity and the significance of employee awareness in minimizing cybersecurity risks.

Efforts on awareness should also be evaluated regularly. Knowledge, reporting, policy compliance, and reaction to controlled exercises are all areas that organisations can explore to reveal whether training ought to be enhanced. Chaudhary, Gkioulos and Katsikas (2022) emphasize that it is crucial to measure cybersecurity awareness programmes instead of relying solely on their implementation.

Support by the organisation is also fundamental since employees might not be willing to adhere to security rules and regulations when they do not know what to expect or do not have the relevant resources. Leadership intervention, good communication and accountability can be used to reinforce secure behaviour (Aksoy, 2024).

These measures are only suggestions relying on literature and are not interventions that were validated by the Python software. The programme can be used to facilitate the organisation and appraisal of publications on training and security behaviour, but the existing validated records are inadequate to prioritise single interventions. Instead, organisations need to choose the right measurements based on their risks and to assess their results instead of expecting that a single training method will be effective in all contexts.

5.6 Theoretical Interpretation Using Protection Motivation Theory and Theory of Planned Behaviour

Protection Motivation Theory (PMT) offers a model to use when thinking about the reaction of employees to perceived threats and their assessment of protective behavior. Employees can be encouraged to adhere to security processes more when they are aware of the severity of a threat and when they are sure that the effect of an attack can be mitigated by taking adequate actions. It can also be the faith they possess in their capability to make such a response. Literature based on PMT to gain insight into the employee cybersecurity behaviour underlines the significance of the protection motivation and responsibility in the responses to cyber threats (Sulaiman et al., 2022).

PMT comes in handy when explaining how awareness programmes may educate on the potential consequences of unsafe behaviour and what effective measures can be undertaken by employees. However, just being aware does not mean safe behaviour. The workers should also be in a position to implement the suggested practices in their work environment. Li, Xu and He (2022) present a review of different factors and mechanisms that can influence cybersecurity protection behaviour.

The Theory of Planned Behaviour (TPB) provides a complementary theory that considers attitudes, perceived social expectations, and perceived ability to control behaviour. Employee attitudes towards security procedures, expectations expressed by their co-workers and supervisors, and trust in behaving in a secure way may influence the intentions of these workers

to behave in a secure way in an organisational context. The study of cybersecurity compliance also reveals that the behaviour of employees can be subject to decision-making styles (Duzenci, Kitapci and Gok, 2023).

A combination of these theories helps to comprehend the relationship between employee awareness and secure behaviour. PMT is more concerned with threat appraisal and utilization of protective responses, TPB on attitudes, social factors and perceived control.

Chapter 6: Conclusion

6.1 Introduction

The chapter is the culmination of the research on human factors in cybersecurity, particularly how employee awareness can be used to address human-induced cyber threats. It determines the achievement of the research aim and objectives, summarises the key findings and explains the contribution of the Python-based research software. The limitations in methodology and software used also contribute to the discussion of the limitations of the interpretation of evidence available. The behaviour of employees and the context of organisations are also crucial factors in comprehending cybersecurity risks (AL-Nuaimi, 2024). Therefore, there is a difference in the conclusions between the evidence found in a scholarly text and the results obtained with the help of the software. Recommendations on how the research workflow can be improved and the practices of employee awareness strengthened are also provided in the chapter. Lastly, possible future research directions are defined to overcome the limitations of the current study and create a more holistic view of the employee cybersecurity behaviour.

6.2 Achievement of the Research Aim and Objectives

The research aimed at revealing the impact that the cybersecurity awareness of employees has on the human-related cybersecurity risks in organisations. The analysis of employee behaviour, phishing vulnerability, awareness programmes, and organisation security practices were analysed using the literature review as a foundation. The different personal and situational factors that influence human cybersecurity behaviour make the consideration of employees into their organisational context important (Alsharida et al., 2023).

The Python software helped in the research process to organize literature records, screening decisions, categorize candidates based on matching keywords, and document evidence when undergoing a manual review. It also provided a way of systematically distinguishing between tentative classifications and confirmed outcomes. However, the effectiveness of employee awareness was not dictated by the software, neither was it established that training reduced cyber incidents.

The research objectives thus were fulfilled in a partial way in terms of literature organisation and carrying out the review process. To make comprehensive conclusions, further confirmations

would be required. In particular, the inconsistency of record information, and the absence of review records, constrain the range of what would be possible to offer by the existing outputs in terms of supporting all research objectives.

6.3 Summary of Key Findings

Literature used in this paper suggests that employee awareness, human behaviour, phishing vulnerability, awareness programmes and organisational security culture are applicable to the knowledge of cybersecurity threats. The human factors have to do with how the workers view the threats, take security decisions, and follow organisational practices (Edeh, 2023). Knowledge related to security and the desire to act can be constructed under the support of awareness and education programmes whose outcomes must be measured respectively (Hu, Hsu and Zhou, 2022).

The software generated results prove the improvements in the organisation and analysis of the corresponding records. The research data file contains eight records in its data set and two publications with labels are contained in the literature register. The confirmed review file contains 2 entries where P001 is fully classified as Phishing Susceptibility and P002 is yet to be fully classified. P001 has review notes that suggest full-text examination prior to its classification being finalised (Hillman, Harel and Toch, 2023).

These products demonstrate the value of systematized record management, but not an overall confirmation of all the eight research records. The differences between the dates of the publications of the topics and description of the topics also should be eliminated. Therefore, the conclusions concerning the general research question rely mainly on the scholarly literature, whereas the outcomes of using the software show the present level of data processing and analysis.

6.4 Contributions to Knowledge and Software Development

The study is useful in the synthesis of the evidence on the problem of employee awareness and human related cybersecurity risks. It fills the gap between the literature-based discussion and a systematic Python workflow that is used to arrange research records, impose rules of key-matching, help with manual verification, and generate outputs that can be inspected.

The difference between provisional and verified candidate classifications is also an important aspect of the workflow. This division gives a system of recording evidence and identifying records which need further scrutiny. The software is therefore a matter of transparency in research and methodical upkeep of literature rather than arbitrarily determining the validity of scholarly findings.

The approach also focuses on the necessity to pay attention to human and organisational aspects and merge them with technical cybersecurity (Aksoy, 2023). It is, however, tentative as the input files are not yet fully reconciled and there is only one record that has been verified to have a valid classification. Further standardisation needs to be made in the data fields, data source trace, and universal eligibility of all the scripts.

6.5 Research and Software Limitations

The research is based on secondary literature and thus, on the relevance, quality and completeness of published evidence. It does not involve a face-to-face data collection among employees or organisations which limits its ability to establish the impact of awareness on behaviour in a given working environment. Their training methods and behavioural outcomes can also be hard to compare due to the differences between studies.

There are other weaknesses with the software. It is possible to find the other relevant terminologies using a keyword matching but may not necessarily reflect on the meaning or context of a publication. Manual verification is incomplete, with P001 in need of further full-text verification and P002 yet to be performed. The cybersecurity studies highlight the significance of consideration of the context in interpreting security-related decisions (Duzenci, Kitapci and Gok, 2023).

P001 and P002 have inconsistencies in the number of years of publication and description of topics in the uploaded files. P003-P008 should be checked in terms of bibliographic information and evidence. The homogeneity of the script inputs and processing rules can also affect the homogeneity of the output. These limitations undermine inferences that can be drawn based on the existing data. The current software results are, therefore, to be viewed as tentative and not as comprehensive as declaring consciousness amongst the workforce or lowering cybersecurity danger.

6.6 Recommendations

The initial suggestion is to finish the process of manual review of pending publications and study the entire text of records whose evidence is not entirely available. The literature register, research data, and confirmed review file ought to be aligned with the year of publication, the description of the topic, identifiers, and bibliography. Any classifications should have evidence traceable and a documented review status.

Software workflow must also be standardised. All the scripts should use the same input paths and field names and the same eligibility rules. The same reconciled evidence base ought to be subject to summary and visualisation scripts, which should be verified with the original records. A preliminary organisation step should involve a classification of key words and verification should be done by the researcher before the publication of conclusions as valid findings.

At organisational level, regular, role-relevant awareness exercises, practical phishing exercises, explicit incident-reporting processes, and appropriate management support should be provided by employers. The behavioural indicators should be used in assessment of the awareness programmes and not attendance. Significance of measurement tools also involves the fact that it is possible to identify whether cybersecurity education has the intended effects (Chaudhary, Gkioulos and Katsikas, 2022). These are recommended solutions that are informed by the literature and not experimentally exercised by Python software.

6.7 Future Research

Future studies ought to implement the enhanced workflow on a bigger, systematically reported set of cybersecurity articles. Future studies could compare the evidence within the organisational industries and discuss how training design, leadership, and security culture could influence the behaviour of employees. The existing secondary research could be enriched with primary research, whereby the employees or cybersecurity professionals will be interviewed, and evidence of the real security practices will be gathered. The software could also be compared with records that are checked manually to establish the accuracy of classification, consistency and completeness. Clear verification criteria and reconciled data set must be established before general conclusions can be drawn about the relationship between employee awareness and mitigating cybersecurity risks.

References

Abrahams, T.O., Farayola, O.A., Kaggwa, S., Uwaoma, P.U., Hassan, A.O. and Dawodu, S.O., 2024. Cybersecurity awareness and education programs: a review of employee engagement and accountability. *Computer Science & IT Research Journal*, *5*(1), pp.100-119.

Aksoy, C., 2023. Critical success factors for cybersecurity just technical? Exploring the role of human factors in cybersecurity management. *Research Journal of Business and Management*, *10*(2), pp.51-57.

Aksoy, C., 2024. Building a cyber security culture for resilient organizations against cyber attacks. *İşletme Ekonomi ve Yönetim Araştırmaları Dergisi*, *7*(1), pp.96-110.

Aljumaiah, O., Jiang, W., Addula, S.R. and Almaiah, M.A., 2025. Analyzing cybersecurity risks and threats in IT infrastructure based on NIST framework. *J. Cyber Secur. Risk Audit*, *2025*(2), pp.12-26.

AL-Nuaimi, M.N., 2024. Human and contextual factors influencing cyber-security in organizations, and implications for higher education institutions: a systematic review. *Global Knowledge, Memory and Communication*, *73*(1-2), pp.1-23.

Alsharida, R.A., Al-rimy, B.A.S., Al-Emran, M. and Zainal, A., 2023. A systematic review of multi perspectives on human cybersecurity behavior. *Technology in society*, *73*, p.102258.

AlSobeh, A.M.R., AlAzzam, I., Shatnawi, A.M.J. and Khasawneh, I., 2023. Cybersecurity awareness factors among adolescents in Jordan: Mediation effect of cyber scale and personal factors. *Online Journal of Communication and Media Technologies*, *13*(2), p.e202312.

Argyridou, E., Nifakos, S., Laoudias, C., Panda, S., Panaousis, E., Chandramouli, K., Navarro-Llobet, D., Mora Zamorano, J., Papachristou, P. and Bonacina, S., 2023. Cyber Hygiene Methodology for Raising Cybersecurity and Data Privacy Awareness in Health Care Organizations: Concept Study. *Journal of Medical Internet Research*, *25*, pp.e41294-e41294.

Aschwanden, R., Messner, C., Höchli, B. and Holenweger, G., 2024. Employee behavior: the psychological gateway for cyberattacks. *Organizational Cybersecurity Journal: Practice, Process and People*, *4*(1), pp.32-50.

Ayodeji, A., Mohamed, M., Li, L., Di Buono, A., Pierce, I. and Ahmed, H., 2023. Cyber security in the nuclear industry: A closer look at digital control systems, networks and human factors. *Progress in Nuclear Energy*, *161*, p.104738.

Azmat, H., 2025. Cybersecurity in Supply Chains: Protecting Against Risks and Addressing Vulnerabilities. *International Journal of Digital Innovation*, *6*(1).

Bernsmed, K., Meland, P.H., Stelkens-Kobsch, T.H., Tedeschi, A., Dambra, C., Buselli, I., Frumento, E., Martintoni, D., Senni, V., Gurtov, A. and Singh Gaba, G., 2024, November. SEC-AIRSPACE: addressing cyber security challenges in future air traffic management. In *SECURWARE 2024: The Eighteenth International Conference on Emerging Security Information, Systems and Technologies, Nov 3-7 2024, Nice, France* (pp. 165-171). International Academy, Research and Industry Association (IARIA).

Burton, S.L., Burrell, D.N., Nobles, C. and Jones, L.A., 2023. Exploring the nexus of cybersecurity leadership, human factors, emotional intelligence, innovative work behavior, and critical leadership traits. *Scientific Bulletin*, *28*(2), pp.162-175.

Chaudhary, S., Gkioulos, V. and Katsikas, S., 2022. Developing metrics to assess the effectiveness of cybersecurity awareness program. *Journal of Cybersecurity*, *8*(1), p.tyac006.

Dalal, A., 2023. Building comprehensive cybersecurity policies to protect sensitive data in the digital era. *Available at SSRN 5424094*.

Delso-Vicente, A.T., Diaz-Marcos, L., Aguado-Tevar, O. and de Blanes-Sebastián, M.G., 2025. Factors influencing employee compliance with information security policies: a systematic literature review of behavioral and technological aspects in cybersecurity. *Future Business Journal*, *11*(1), p.28.

Duzenci, A., Kitapci, H. and Gok, M.S., 2023. The role of decision-making styles in shaping cybersecurity compliance behavior. *Applied Sciences*, *13*(15), p.8731.

Edeh, N.C., 2023. Cybersecurity and human factors: A literature review. *Cybersecurity for decision makers*, pp.45-56.

Ejiofor, O., Obu, A., Yusuf, T., Tonui, I. and Yusuf, B., 2025. Human factors in cybersecurity: Training and awareness for analysts. *Comput. Sci. IT Res. J*, *6*, pp.252-265.

Georgiadou, A., Mouzakitis, S. and Askounis, D., 2022. Detecting insider threat via a cyber-security culture framework. *Journal of Computer Information Systems*, *62*(4), pp.706-716.

Georgiadou, A., Mouzakitis, S., Bounas, K. and Askounis, D., 2022. A cyber-security culture framework for assessing organization readiness. *Journal of Computer Information Systems*, *62*(3), pp.452-462.

Hakimi, M., Quchi, M.M. and Fazil, A.W., 2024. Human factors in cybersecurity: an in depth analysis of user centric studies. *Jurnal Ilmiah Multidisiplin Indonesia (JIM-ID)*, *3*(01), pp.20-33.

Hassandoust, F. and Johnston, A.C., 2023. Peering through the lens of high-reliability theory: a competencies driven security culture model of high-reliability organisations. *Information Systems Journal*, *33*(5), pp.1212-1238.

Hijji, M. and Alam, G., 2022. Cybersecurity awareness and training (CAT) framework for remote working employees. *Sensors*, *22*(22), p.8663.

Hu, S., Hsu, C. and Zhou, Z., 2022. Security education, training, and awareness programs: Literature review. *Journal of Computer Information Systems*, *62*(4), pp.752-764.

Jabir, R., Le, J. and Nguyen, C., 2025. Phishing Attacks in the Age of Generative Artificial Intelligence: A Systematic Review of Human Factors. *AI*, *6*(8), p.174.

Jones, L.A., 2024. Unveiling human factors: Aligning facets of cybersecurity leadership, insider threats, and arsonist attributes to reduce cyber risk. *SocioEconomic Challenges*, *8*(2), pp.44-63.

Kelley, S., 2022. Employee perceptions of the effective adoption of AI principles. *Journal of Business Ethics*, *178*(4), pp.871-893.

Khadka, K. and Ullah, A.B., 2025. Human factors in cybersecurity: an interdisciplinary review and framework proposal: K. Khadka, AB Ullah. *International Journal of Information Security*, *24*(3), p.119.

Kioskli, K., Fotis, T., Nifakos, S. and Mouratidis, H., 2023. The importance of conceptualising the human-centric approach in maintaining and promoting cybersecurity-hygiene in healthcare 4.0. *Applied Sciences*, *13*(6), p.3410.

Klein, G. and Zwilling, M., 2024. The weakest link: employee cyber-defense behaviors while working from home. *Journal of Computer Information Systems*, *64*(3), pp.408-422.

Kont, K.R., 2024. Libraries and cyber security: the importance of the human factor in preventing cyber attacks. *Library Hi Tech News*, *41*(1), pp.11-15.

Li, L., Xu, L. and He, W., 2022. The effects of antecedents and mediating factors on cybersecurity protection behavior. *Computers in Human Behavior Reports*, *5*, p.100165.

Mayeke, N.R., Arigbabu, A.T., Olaniyi, O.O., Okunleye, O.J. and Adigwe, C.S., 2024. Evolving access control paradigms: A comprehensive multi-dimensional analysis of security risks and system assurance in cyber engineering. *Asian Journal of Research in Computer Science*, *17*(5), pp.108-124.

Mohammad, T., Hussin, N.A.M. and Husin, M.H., 2022. Online safety awareness and human factors: An application of the theory of human ecology. *Technology in Society*, *68*, p.101823.

Nasir, S., 2023, July. Exploring the effectiveness of cybersecurity training programs: factors, best practices, and future directions. In *Proceedings of the Cyber Secure Nigeria Conference* (Vol. 1, pp. 9-22).

Ncubukezi, T., 2022, March. Human errors: A cybersecurity concern and the weakest link to small businesses. In *Proceedings of the 17th International Conference on Information Warfare and Security* (p. 395). Albany, New York, USA: Academic Conferences International Limited.

Nobles, C., 2022. Stress, burnout, and security fatigue in cybersecurity: A human factors problem. *HOLISTICA–Journal of Business and Public Administration*, *13*(1), pp.49-72.

Nurwanah, A., 2024. Cybersecurity in accounting information systems: Challenges and solutions. *Advances in Applied Accounting Research*, *2*(3), pp.157-168.

Nwankpa, J.K. and Datta, P.M., 2023. Remote vigilance: The roles of cyber awareness and cybersecurity policies among remote workers. *Computers & Security*, *130*, p.103266.

Peca, L. and Tsurcanu, D., 2025. Reducing cyber risk through a human–centred approach. *Journal of Engineering Science*, (1), pp.18-31.

Peca, L., Cojocaru, S., Dumitraşcu, M. and Tsurcanu, D., 2025. Evaluation of cybersecurity training perceptions, adopted practices and strategic directions for capacity building. *Journal of Engineering Science*, (3), pp.75-90.

Pollini, A., Callari, T.C., Tedeschi, A., Ruscio, D., Save, L., Chiarugi, F. and Guerri, D., 2022. Leveraging human factors in cybersecurity: an integrated methodological approach. *Cognition, Technology & Work*, *24*(2), pp.371-390.

Prabhu, S. and Thompson, N., 2022. A primer on insider threats in cybersecurity. *Information Security Journal: A Global Perspective*, *31*(5), pp.602-611.

Qureshi, R. and Koo, I., 2026. A comprehensive survey of cybersecurity threats and data privacy issues in healthcare systems. *Applied Sciences*, *16*(3), p.1511.

Rongala, S., 2025. Best practices in cybersecurity training: Enhancing employee awareness to mitigate risks. *International Journal of Scientific Research in Computer Science, Engineering and Information Technology*, *11*(1), pp.538-547.

Safitra, M.F., Lubis, M. and Fakhrurroja, H., 2023. Counterattacking cyber threats: A framework for the future of cybersecurity. *Sustainability*, *15*(18), p.13369.

Sangwan, A., 2024, May. Human factors in cybersecurity awareness. In *2024 International Conference on Intelligent Systems for Cybersecurity (ISCS)* (pp. 1-7). IEEE.

Shah, M.U., Iqbal, F., Rehman, U. and Hung, P.C., 2023. A comparative assessment of human factors in cybersecurity: Implications for cyber governance. *IEEE Access*, *11*, pp.87970-87984.

Sulaiman, N.S., Fauzi, M.A., Hussain, S. and Wider, W., 2022. Cybersecurity behavior among government employees: The role of protection motivation theory and responsibility in mitigating cyberattacks. *Information*, *13*(9), p.413.

Taherdoost, H., 2024. A critical review on cybersecurity awareness frameworks and training models. *Procedia computer science*, *235*, pp.1649-1663.

Thron, E., Faily, S., Dogan, H. and Freer, M., 2024. Human factors and cyber-security risks on the railway–the critical role played by signalling operations. *Information & Computer Security*, *32*(2), pp.236-263.

Tikanmäki, I. and Ruoslahti, H., 2024. Insights on Human Factors Enhancing Cybersecurity. *Information & Security: An International Journal*.

Tolossa, D., 2023. Importance of cybersecurity awareness training for employees in business. *Vidya-A Journal of Gujarat University*, *2*(2), pp.104-107.

Triplett, W.J., 2022. Addressing human factors in cybersecurity leadership. *Journal of Cybersecurity and Privacy*, *2*(3), pp.573-586.

Ugbebor, F., Aina, O., Abass, M. and Kushanu, D., 2024. Employee cybersecurity awareness training programs customized for SME contexts to reduce human-error related security incidents. *Journal of Knowledge Learning and Science Technology ISSN: 2959-6386 (online)*, *3*(3), pp.382-409.

Waddell, M., 2024, January. Human factors in cybersecurity: Designing an effective cybersecurity education program for healthcare staff. In *Healthcare management forum* (Vol. 37, No. 1, pp. 13-16). Sage CA: Los Angeles, CA: SAGE Publications.

Willie, M.M., 2023. The role of organizational culture in cybersecurity: building a security-first culture. *Journal of Research, Innovation and Technologies*, *2*(2 (4)), pp.179-198.

Yeng, P.K., Fauzi, M.A. and Yang, B., 2022. A comprehensive assessment of human factors in cyber security compliance toward enhancing the security practice of healthcare staff in paperless hospitals. *Information*, *13*(7), p.335.

Zangana, H.M., Sallow, Z.B. and Omar, M., 2025. The human factor in cybersecurity: Addressing the risks of insider threats. *Jurnal Ilmiah Computer Science*, *3*(2), pp.76-85.

Zureigat, B.N., Alhanatleh, H., Alghizzawi, M. and Alrifae, A.A.M., 2025. Role of digital technologies based on cybersecurity awareness in enhancing firms' performance: digital accounting systems setting. *Economics*, *13*(3), pp.375-396.
Grandhi, S.R. and Still, J.D., 2025, September. Deciphering human error: Improving cybersecurity reporting. In *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* (Vol. 69, No. 1, pp. 234-239). Sage CA: Los Angeles, CA: SAGE Publications.

Websites

nist.gov, 2026. *Information Technology Laboratory*. Available from:

https://www.nist.gov/itl/smallbusinesscyber/guidance-topic [Accessed on: 17.08.2026]

guides.himmelfarb.gwu.edu, 2026. *Study Design 101: Meta-Analysis*. Available from: https://guides.himmelfarb.gwu.edu/studydesign101/metaanalysis [Accessed on: 17.08.2026]

cyberireland.ie, 2026. *Key factors in Human Behaviour for Cyber-Security*. Available from: https://cyberireland.ie/key-factors-in-human-behaviour-for-cyber-security/ [Accessed on: 17.08.2026]

adnovum.com, 2026. *Creating a Cybersecurity Culture in the Workplace.* Available from: https://www.adnovum.com/blog/cybersecurity-culture [Accessed on: 17.08.2026]

expertprogrammanagement.com, 2026. *Theory of Planned Behavior.* Available from: https://expertprogrammanagement.com/2019/12/theory-of-planned-behavior/[Accessed on: 17.08.2026]

arysturt.free- online.co.uk, 2026. *Theories of Health Behaviour.* Available from: http://www.garysturt.free-online.co.uk/health/healat2.htm [Accessed on: 17.08.2026]

researchgate.net, 2026. *Research Approach (research Onion).* Available from: https://www.researchgate.net/figure/Saunders-et-als-Research-Onion-30_fig2_349316382 [Accessed on: 17.08.2026]

threatcop.com, 2026. *Human Errors in Cybersecurity.* Available from: https://threatcop.com/blog/top-5-cyber-attacks-and-security-breaches-due-to-human-error/ [Accessed on: 17.08.2026]

fastercapital.com, 2026. *Employee Training And Awareness Programs.* Available from: https://fastercapital.com/topics/employee-training-and-awareness-programs.html/1 [Accessed on: 17.08.2026]

secureitworld.com, 2026. *Making Security Everyone's Job – The 7 Key Dimensions of Security Culture.* Available from: https://www.secureitworld.com/blog/making-security-everyones-job-the-7-key-dimensions-of-security-culture/ [Accessed on: 17.08.2026]

Appendix

Literature Review and Thematic Analysis Matrix

No.	Author(s) and Year	Main Focus	Key Finding Relevant to Study	Theme
1	Duzenci, Kitapci and Gok (2023)	Cybersecurity compliance behaviour	Decision-making styles influence employees' cybersecurity compliance behaviour.	Theme 1: Human Error
2	Edeh (2023)	Human factors in cybersecurity	Human factors represent an important dimension of organisational cybersecurity.	Theme 1: Human Error
3	Ejiofor *et al.* (2025)	Training and awareness	Training and awareness are important for addressing human factors in cybersecurity.	Theme 2: Awareness Programmes
4	Georgiadou, Mouzakitis and Askounis (2022)	Cybersecurity culture and insider threats	Security culture can support organisational readiness and the detection of insider threats.	Theme 3: Security Culture
5	Hakimi, Quchi and Fazil (2024)	User-centric cybersecurity	Cybersecurity research needs to consider user behaviour and human factors.	Theme 1: Human Error
6	Hijji and Alam (2022)	Remote-working awareness	Remote workers require specific cybersecurity awareness and training approaches.	Theme 2: Awareness Programmes

7	Jabir, Le and Nguyen (2025)	AI-enabled phishing	Generative AI is contributing to increasingly sophisticated phishing threats targeting human behaviour.	Theme 1: Human Error
8	Jones (2024)	Cybersecurity leadership and insider threats	Leadership and human factors are important considerations in reducing cyber risk.	Theme 3: Security Culture
9	Khadka and Ullah (2025)	Human factors framework	Cybersecurity requires an interdisciplinary understanding of human factors.	Theme 1: Human Error
10	Kioskli *et al.* (2023)	Human-centred cybersecurity hygiene	A human-centric approach is important for maintaining cybersecurity hygiene.	Theme 2: Awareness Programmes